SECTION-I (i) (Maximum Marks:15)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If ABCD is regular tetrahedron with length of any edge 'ℓ', then the minimum distance of any vertex from the opposite face is -

(A) $\sqrt{\frac{2}{3}}\ell$ (B) $\sqrt{\frac{3}{2}}\ell$

(C) $\frac{2}{3}\ell^2$ (D) $\frac{1}{\sqrt{3}}\ell$

2. Two lines whose equations are $L_1 : \frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{\lambda}$ and $L_2 : \frac{x-2}{3} = \frac{y-3}{2} = \frac{z-2}{3}$ lie in the same plane. If L_1 intersects a plane $x + y + z = 15$ at P, then distance of P from (3, 4, 3) is -

(A) 6 (B) 4

(C) 2 (D) 3

3. Vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar vectors such that $|\vec{a}| = 2, |\vec{b}| = 3$ & $\vec{a} \wedge \vec{b} = 30°$. A unit vector $\vec{d}$ is perpendicular to all of them. If $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \frac{\hat{i}}{6} - \frac{\hat{j}}{3} + \frac{\hat{k}}{3}$, then $|\vec{c}.\hat{i}| + |\vec{c}.\hat{j}| + |\vec{c}.\hat{k}|$ is equal to -

(A) $\frac{5}{3}$ (B) $\frac{5}{9}$ (C) $\frac{5}{12}$ (D) $\frac{5}{18}$

4. If two diagonals of two rhombus are represented by vectors $\vec{d}_1 = \vec{a} + 2\vec{b}$, $\vec{d}_2 = 2\vec{a} - \vec{b}$ and $\vec{d}_3 = 2\vec{a} + \vec{b}$, $\vec{d}_4 = \vec{a} - 2\vec{b}$respectively, then $\left[\vec{a}\ \vec{b}\ \vec{a} \times 2\vec{b}\right]$equals :-

(A) $\frac{|\vec{a}|^2|\vec{b}|^2}{4}$ (B) $\frac{|\vec{a}|^2|\vec{b}|^2}{2}$

(C) $2|\vec{a}|^2|\vec{b}|^2$ (D) $4|\vec{a}|^2|\vec{b}|^2$

5. Let $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$ and $\vec{c} = \alpha\vec{a} + \beta\vec{b}$. If the vectors, $\hat{i} - 2\hat{j} + \hat{k}$, $3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c}$ are coplanar then $\frac{\alpha}{\beta}$ is -

(A) 1 (B) 2 (C) 3 (D) – 3

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SECTION-I (ii) : (Maximum Marks: 20)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

6. A plane meets the coordinate axes in A, B, C such that the centroid of the triangle ABC is the point $(1, r, r^2)$. The plane passes through the point $(4, -8, 15)$ if r is equal to -

(A) -3 (B) 3

(C) 5 (D) -5

7. A non-zero vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j}, \hat{i}-\hat{k}$,. The possible angle between $\vec{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$ is -

(A) $\pi/3$ (B) $\pi/4$

(C) $\pi/6$ (D) $3\pi/4$

8. If unit vectors $\hat{i}$ & $\hat{j}$ are at right angles to each other and $\vec{p} = 3\hat{i} + 4\hat{j}$, $\vec{q} = 5\hat{i}$, $4\vec{r} = \vec{p} + \vec{q}$ and $2\vec{s} = \vec{p} - \vec{q}$, then -

(A) $|\vec{r} + k\vec{s}| = |\vec{r} - k\vec{s}|$ for all real k (B) $\vec{r}$ is perpendicular to $\vec{s}$

(C) $\vec{r} + \vec{s}$ is perpendicular to $\vec{r} - \vec{s}$ (D) $|\vec{r}| = |\vec{s}| = |\vec{p}| = |\vec{q}|$

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9. If a vector $\vec{r}$ of magnitude $3\sqrt{6}$ is directed along the bisector of the angle between the vectors $\vec{a} = 7\hat{i} - 4\hat{j} - 4\hat{k}$ & $\vec{b} = -2\hat{i} - \hat{j} + 2\hat{k}$, then $\vec{r}$ is equal to -

(A) $\hat{i} - 7\hat{j} + 2\hat{k}$ (B) $\hat{i} + 7\hat{j} - 2\hat{k}$

(C) $-\hat{i} + 7\hat{j} - 2\hat{k}$ (D) $\hat{i} - 7\hat{j} - 2\hat{k}$

10. A line passes through two points A and B whose position vectors are $\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+\hat{k}$ respectively. The position vector of a point on it at a unit distance from the point A is -

(A) $\frac{1}{5}(5\hat{i}+\hat{j}-7\hat{k})$ (B) $\frac{1}{5}(4\hat{i}+9\hat{j}-13\hat{k})$

(C) $\frac{1}{5}(6\hat{i}+\hat{j}-7\hat{k})$ (D) $\frac{1}{5}(5\hat{i}+9\hat{j}-13\hat{k})$

SECTION-I (iii) : (Maximum Marks: 9)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Questions 11 - 13

A plane containing the lines $L_1 : 2x - y + 3z - 5 = 0 = x - 2y + 2z + 1$ and $L_2 : \frac{x-1}{\lambda-1} = \frac{y-2}{\lambda-2} = \frac{z-3}{\lambda-3}$.

If P be a variable point on the plane which is at a distance 4 from (0, 0, 0).

On the basis of above information answer the following :

11. Equation of the plane is -

(A) $x + y + z = 6$ (B) $2x - y + 3z - 9 = 0$

(C) $x - 2y + 2z - 3 = 0$ (D) can't be determined

12. The possible value of λ is -

(A) 0 (B) 1

(C) 2 (D) 3

13. Area of the region bounded by the locus of P is -

(A) π (B) 2π

(C) 4π (D) 9π

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SECTION-II (ii) : (Maximum Marks: 20)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors such that $|\vec{a}| = \sqrt{6}$, $|\vec{b}| = \sqrt{2}$, $|\vec{c}| = \sqrt{3}$ and a vector $\vec{v} = (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} + (\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} + (\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b}$ where magnitude of $\vec{v}$ is M then $\frac{M^2}{100}$ is

2. Consider a line passing through the point P(3,4,5) & having direction ratios (2,–2,1). If image of point Q(1,–1,8) w.r.t. given line is Q', then length of median of $\Delta PQQ'$ through vertex P is -

3. If lines $bx = 1 - cy$, $bz = 1$ & $bx = 2 + ay$, $bz = 3 + cy$ are perpendicular, where a,b,c are sides of ΔABC, then minimum value of cos B is -

4. Let $\hat{a} + 2\hat{b} + \hat{c} = \hat{a} \times \hat{c}$ then $|\hat{a} \times \hat{b} + \hat{b} \times \hat{c} + \hat{a} + \hat{b} + \hat{c}|$ is equal to

5. Let $|\vec{a}| = 4$, $|\vec{b}| = 1$ and $\vec{b} \times \vec{\alpha} + \vec{a} = \vec{\alpha}$. If projection of $\vec{\alpha}$ along $\vec{b}$ is 3 then projection of $\vec{\alpha}$ along $\vec{a}$ is

SECTION–IV : (Maximum Marks: 8)

Matrix Match Type (One or More than one option correct)

For each entry in Column-I , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		Column-II
(A)	Vectors $\vec{a},\vec{b},\vec{c}$ are of same magnitude when taken pairwise they makes equal angle. If $\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=\hat{i}+\hat{k}$, then $\vec{c}$ is	(P)	(6, 16, 20)
(B)	If vector $\vec{r}$ satisfies $\vec{r}\times(2\hat{i}+\hat{j}+\hat{k})=\hat{j}-\hat{k}$, then $\vec{r}$ can be	(Q)	(0, 1, 1)
(C)	If plane passes through (3, 4, 5) and meets coordinates axes in A, B, C, then point of intersection of plane through A, B, C parallel to coordinate planes can be	(R)	(3, 1, –1)
(D)	$\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=2\hat{i}-\hat{k}$, then point of intersection of line $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$ and $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ is	(S)	$\left(\frac{4}{3},-\frac{1}{3},-\frac{1}{3}\right)$
		(T)	(3, 2, 2)

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Section–I(i) Single Correct Type (Maximum Marks:30)

1. Let $\vec{x}$, $\vec{y}$ are non zero and non collinear vectors such that

$(a\alpha^2 + b\alpha + c)\vec{x} + (a\beta^2 + b\beta + c)\vec{y} + (a\gamma^2 + b\gamma + c)(\vec{x} \times \vec{y}) = 0$, then $(a + b + c)$ is (where $a,b,c \in R$)

(A) -2 (B) -1 (C) 0 (D) 4

2. If for unit vectors $\hat{a}, \hat{b}$ and non-zero $\vec{c}$, $\hat{a} \times \hat{b} + \hat{a} = \vec{c}$ and $\hat{b}.\vec{c} = 0$, then volume of parallelopiped with coterminous edges $\hat{a}, \hat{b}$ and $\vec{c}$ will be (in cu.units)-

(A) 6 (B) 4 (C) 1 (D) $\frac{1}{2}$

3. If $\vec{d} = \lambda\left(\vec{a} \times \vec{b}\right) + \mu\left(\vec{b} \times \vec{c}\right) + v(\vec{c} \times \vec{a})$

$\left[\vec{a} \quad \vec{b} \quad \vec{c}\right] = \frac{1}{8}$ and $\vec{d}.\left(\vec{a} + \vec{b} + \vec{c}\right) = 8$, then $\lambda + \mu + v$ is

(A) 32 (B) 64 (C) 27 (D) 19

4. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{W} = \hat{i} + 3\hat{k}$, $|\vec{U}| = 2$. If $\vec{U}$ is a vector in x-y plane, then greatest value of $\left([\vec{U}\ \vec{V}\ \vec{W}]\right)^2$ is -

(A) 232 (B) 340 (C) 236 (D) 312

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5. In a tetrahedron OABC, the edges are of lengths, $|OA| = |BC| = a$, $|OB| = |AC| = b$, $|OC| = |AB| = c$. Let G_1 and G_2 be the centroids of the triangle ABC and AOC such that $OG_1 \perp BG_2$, then the value of $\frac{a^2 + c^2}{b^2}$ is (O being origin)

(A) 2 (B) 3 (C) 6 (D) 9

6. The x-intercept of the plane which is passing through the intersection of the planes

$x + 2y + 3z + 5 = 0$ and $2x - 3y + 7z + 1 = 0$ and parallel to line $\vec{r} = \hat{i} + 2\hat{j} + \lambda\left(8\hat{i} - 7\hat{j} - 4\hat{k}\right)$, is-

(A) $\frac{13}{5}$ (B) $-\frac{7}{5}$ (C) $\frac{6}{5}$ (D) $-\frac{9}{5}$

7. The direction ratios of a normal to the plane passing through (1,0,0), (0,1,0) and making an angle of $\frac{\pi}{4}$ with plane $x + y = 3$ can be -

(A) 0,1,0 (B) $1, 1, \sqrt{2}$ (C) 1,0,0 (D) $\sqrt{2}, 1, 1$

8. The equation of plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from point (0,0,0) is

(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$

(C) $3x + 4y + 5z = 49$ (D) $x + 7y - 5z = 2$

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9. The direction cosines of the projection of the line $\frac{1}{2}(x-1) = -y = z+2$ on the plane $2x + y - 3z = 4$ are-

(A) $\left(\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ (B) $\left(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

(C) $\left(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$ (D) None of these

10. A unit vector $\hat{a}$ is such that $\hat{a} \times \vec{b} = \hat{i} - \hat{k}$, where $\vec{b}$ is $\hat{i} + \hat{j} + \hat{k}$, then the value of $|\hat{a}.\vec{b}|$ will be-

(A) 1 (B) 2 (C) 3 (D) 4

Section–I(ii) Multiple Correct Type : (Maximum Marks: 24)

11. If $\vec{a} = \hat{i} + 6\hat{j} + 3\hat{k}$; $\vec{b} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = (\alpha+1)\hat{i} + (\beta-1)\hat{j} + \hat{k}$ are linearly dependent vectors & $|\vec{c}| = \sqrt{6}$, then possible value(s) of $(\alpha + \beta)$ can be-

(A) 1 (B) 2 (C) 3 (D) 4

12. Let $a_1, a_2, a_3 \in R - \{0\}$ and $a_1 + a_2\cos 2x + a_3\sin^2 x = 0 \ \forall \ x \in R$, then-

(A) vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = 4\hat{i} + 2\hat{j} + \hat{k}$ are perpendicular

(B) vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} + 2\hat{k}$ are parallel to each other

(C) if vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is of length $\sqrt{6}$, then one of ordered triplet $(a_1, a_2, a_3) = (1, -1, -2)$

(D) if $2a_1 + 3a_2 + 6a_3 = 26$, then $|a_1\hat{i} + a_2\hat{j} + a_3\hat{k}| = 2\sqrt{6}$

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13. ΔABC is obtuse angled at B and points D and E are taken on sides AC and BC respectively such that AD : DC = 2 : 3 and BE : EC = 1 : 2. If DE produced meet AB produced at F, then -

(A) AF : BF = 4 : 3 (B) AF : BF = 3 : 1

(C) DE : EF = 1 : 5 (D) DE : EF = 3 : 4

14. If angle between the vectors $\vec{u} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{v} = 2x\hat{i} + x\hat{j} - \hat{k}$ is acute and the angle between the vector $\vec{v}$ and y-axis is obtuse, then x^2 may take the value -

(A) $\frac{1}{9}$ (B) 1 (C) 2 (D) $\sqrt[5]{3}$

15. Let $\bar{a} \times \bar{b} + \bar{b} = \bar{c} \times \bar{a} + \bar{b} \times \bar{c}$ where $|\bar{a}| = |\bar{b}|$ then identify the correct statement(s)

(A) $\bar{a}.\bar{b} = 0$ (B) $\bar{b}.\bar{c} = 0$

(C) $\bar{c}.\bar{a} = 0$ (D) $\bar{c} = \bar{0}$

16. From point A(−1, 1, −1), the perpendicular is drawn to a plane P_1 whose foot is (1, −1, 1). From point B(3, −1, −1), the perpendicular is drawn to a plane P_2 whose foot is (1, 2, 1). P_3 is a plane containing the line of intersection of P_1 and P_2 and passing through the mid point of the line joining A and B. Then which of the following statement(S) is/are correct?

(A) Direction ratios of line of intersection of P_1 and P_2 are 5, 4, −1.

(B) Equation of plane P_3 is $16x - 19y + 4z = 12$

(C) Equation of line of intersection of P_1 and P_2 is $\frac{x-15}{5} = \frac{y-12}{4} = \frac{z-0}{-1}$

(D) The length of perpendicular from (0, 0, 0) to plane P_3 is $\frac{2}{\sqrt{18}}$

Section–I(iii) Comprehension Type : (Maximum Marks: 18)

Paragraph for Questions 17 and 18

Given $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and L be a line whose equation is $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$.

17. The shortest distance of the point P(0,1,2) from the line L is equal to -

(A) $\frac{\sqrt{119}}{3}$ (B) 2 (C) $\frac{\sqrt{219}}{3}$ (D) $\frac{\sqrt{13}}{3}$

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18. If 'C' be the curve in the x-y plane satisfying $\vec{R}.(2\hat{i} - 2\hat{j} - \vec{R}) = 1$ (where $\vec{R}$ be the position vector of a variable point on the curve), then the minimum distance of line 'L' from the centre of curve 'C' is-

(A) $2\sqrt{3} - 1$ (B) $2\sqrt{3}$ (C) 4 (D) $\sqrt{11} - 1$

Paragraph for Questions 19 and 20

Consider a tetrahedron PQRS with PQ = 4 units. A plane 'P_1' is drawn perpendicular to PQ. Let P',R' & S' denotes the respective foot of perpendiculars drawn from P,R & S to the plane 'P_1' such that area of $\Delta P'R'S'$ = 15 sq. units.

19. Volume of the tetrahedron PQRS (in cu. units) is -

(A) $\frac{47}{2}$ (B) 20 (C) 25 (D) $\frac{57}{2}$

20. If the distance of S' from plane PQR is 5 units then the distance of R from line through PQ is -

(A) 5 units (B) 6 units (C) 8 units (D) 10 units

Paragraph for Questions 21 and 22

Consider a plane P : $x + y + z = 3$ and a circle $x^2 + y^2 = 1, z = 0$

21. If $A(x_1, y_1, 0)$ and $B(x_2, y_2, 0)$ are the coordinates of the points on the circle which are at maximum and minimum distance from the plane P, then $|(x_1 + y_1) - (x_2 + y_2)|$ is equal to

(A) 0 (B) $\sqrt{2}$ (C) $2\sqrt{2}$ (D) 2

22. Image of the points on the circle in the plane P lies on

(A) $x + y - 2z = 1$ (B) $x + y + z = 5$

(C) $2x + y + 2z = 10$ (D) $2x + 2y - z = 6$

Section–I(iv) Matrix Match Type : (Maximum Marks: 6)

23. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	$\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. Given $\vec{x} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{x}.\vec{a} = 0$, then $\vec{x} - \vec{c}$ is	(1)	$\frac{1}{7}\left(5\hat{i} + 17\hat{j}\right)$
(Q)	If $\vec{a} = \hat{i} + 2\hat{j}$, $\vec{b} = 2\hat{i} - \hat{k}$, $\vec{x} \times \vec{a} = \vec{b} \times \vec{a}$, $\vec{x} \times \vec{b} = \vec{a} \times \vec{b}$, then $\sqrt{14}\frac{\vec{x}}{\lvert\vec{x}\rvert}$ is	(2)	$\hat{i} - \frac{7}{5}\hat{j} + \frac{1}{5}\hat{k}$
(R)	A line passes through the points A and B whose position vectors are $\hat{i} - 2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - 3\hat{k}$ respectively. The position vector of a point lying on the line segment AB at unit distance from point A is	(3)	$3\hat{i} + 2\hat{j} - \hat{k}$
(S)	The position vectors of two vertices of a triangle are $-\hat{i} + 3\hat{j}$ and $2\hat{i} + 5\hat{j}$ and orthocentre is $\hat{i} + 2\hat{j}$,then the third vertex is	(4)	$-5\left(\hat{i} + \hat{j} + \hat{k}\right)$

(A) P ⟶ 4;Q ⟶ 3;R ⟶ 2;S ⟶ 1

(B) P ⟶ 4;Q ⟶ 3;R ⟶ 1;S ⟶ 2

(C) P ⟶ 3;Q ⟶ 4;R ⟶ 2;S ⟶ 1

(D) P ⟶ 3;Q ⟶ 4;R ⟶ 1;S ⟶ 2

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24.

	List-I		List-II
(P)	Let volume of a tetrahedron ABCD is $\frac{81}{2}$ cube unit and volume of parallelopiped whose three coterminous edges are line segments joining centroid of any face of tetrahedron with centroids of its other three faces is V cubic unit, then V is	(1)	9
(Q)	If image of the point (1, 0, 1) in the plane $x - y - z = 1$ is (a,b,c) then $3(a - b + c)$ is equal to	(2)	8
(R)	Locus of all the points which are at a distance of 3 units from the line $\vec{r} = \lambda(i + j + k)$ is given by $x^2 + y^2 + z^2 - xy - yz - zx = \frac{k}{2}$, then find $\frac{k}{9}$	(3)	3
(S)	$\vec{a}, \vec{b}$ and $\vec{c}$ are such that $\lvert\vec{a}\rvert = \sqrt{3}, \lvert\vec{b}\rvert = 2$ and $\lvert\vec{c}\rvert = \sqrt{6}$. If $\vec{a}.\vec{b} < 0; \vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{a}, \vec{b}, \vec{c}$ form the coterminous edges of a tetrahedron of unit volume, then angle between $\vec{a}$ and $\vec{b}$ is $\frac{p\pi}{q}$, (where p, q are coprime numbers), where (p + q) is	(4)	7

(A) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4 (B) P ⟶ 4;Q ⟶ 3;R ⟶ 2;S ⟶ 1

(C) P ⟶ 1;Q ⟶ 4;R ⟶ 3;S ⟶ 2 (D) P ⟶ 1;Q ⟶ 2;R ⟶ 4;S ⟶ 3

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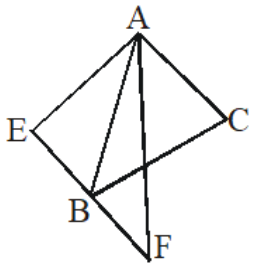
Section–II (iii) Numerical Type:(Upto Second Decimal place) (Maximum Marks: 24)

1. Let $\vec{r} = \sin x\left(\vec{a} \times \vec{b}\right) + \cos y\left(\vec{b} \times \vec{c}\right) + 2(\vec{c} \times \vec{a})$, where $\vec{a}, \vec{b}$ & $\vec{c}$ are non-zero and non-coplanar vectors. If $\vec{r}$ is orthogonal to $\vec{a} + \vec{b} + \vec{c}$, then minimum value of $\frac{(x^2 + y^2)}{\pi^2}$ is

2. A unit vector $\hat{a}$ is such that $\hat{a} \times \vec{b} = \hat{i} - \hat{k}$, where $\vec{b}$ is $\hat{i} + \hat{j} + \hat{k}$. Value of $\left|\hat{a}.\vec{b}\right|$ will be-

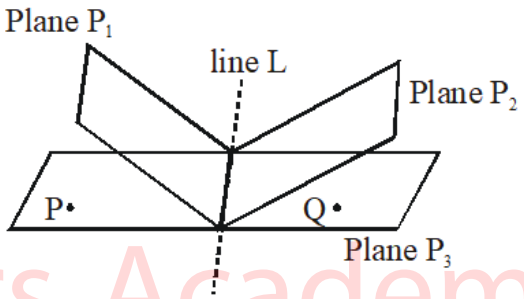
3. If ΔABC and ΔAEF are such that EB = BF, AB = EF = 1, BC = 6, AC = $\sqrt{33}$ and $\overrightarrow{AB}.\overrightarrow{AE} + \overrightarrow{AC}.\overrightarrow{AF} = 2$, then the value of $\overrightarrow{BC}.\overrightarrow{BF}$ is (given figure)



4. A plane passes point P(0, 1, −1) and contains a line L : $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z+2}{1}$. If R is a point on line L such that ΔPQR is right angled at P, where Q(1, 1, −2), then the value of $\frac{PR^2}{5}$ is

5. In a tetrahedron LMNO edges ML, MN & MO are mutually perpendicular and length of altitudes drawn from O, L and N to opposite faces are 1,2 and 3 units respectively, then length of altitude drawn from M to face LNO is -

6. In adjacent diagram equation of plane P_1 and P_2 are $x + y - 2z = 1$ and $ax - y + z = 2$ respectively (where $\alpha \in Q$). Coordinates of points P and Q are $(\alpha,1,3)$ and $\left(\frac{5}{3}, 1, \alpha\right)$ respectively on the plane P_3. if ℓ,m,n are direction cosines of line L then absolute value of $\frac{\sqrt{35}(\ell + m + n)}{\alpha}$ is



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Important Notes

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Section–I(i) (Maximum Marks:24)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. All roots of the equation, $(1 + z)^6 + z^6 = 0$ -

(A) lie on a unit circle with centre at the origin

(B) lie on a unit circle with centre at $(-1, 0)$

(C) lie on the vertices of a regular polygon with centre at the origin.

(D) are collinear

2. If $P = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$, $a = P + P^2 + P^4$ and $b = P^3 + P^5 + P^6$, then the value of $|(a + b) + abi|$ (where $i = \sqrt{-1}$) is -

(A) $\sqrt{3}$ (B) $\sqrt{5}$ (C) $\sqrt{6}$ (D) 2

3. If z_1, z_2, z_3 are the roots of the equation $z^3 - z^2(1 + 3i) + z(3i - 2) + 2 = 0$, then $\text{Im}(z_1) + \text{Im}(z_2) + \text{Im}(z_3)$ is -

(A) 0 (B) 1 (C) 2 (D) 3

4. Number of values of z satisfying $|z - 2i| = 2$ and $z(1 - i) - \bar{z}(1 + i) = 4i$ simultaneously is -

(A) 0 (B) 2 (C) 3 (D) 4

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5. Value of $\sum_{k=1}^{100}\left(i^{k!} + \omega^{k!}\right)$, where $i = \sqrt{-1}$ and ω is complex cube root of unity, is -

(A) $190 + \omega$ (B) $192 + \omega^2$ (C) $190 + i$ (D) $192 + i$

6. If z_2 & z_4 are conjugates of z_1 & z_3 respectively, then $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ is equal to -

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) π (D) 0

7. If $|z - 3 - 2i| = |z + 2i|$, where z is a complex number, then minimum value of $|z|$ will be -

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$ (C) $\frac{7}{10}$ (D) $\frac{9}{10}$

8. If one of the vertex of the regular hexagon, circumscribing the circle $|z - 1 - i| = \sqrt{3}$ is $2 + (1 + \sqrt{3})i$, then the complex number which is not representing any vertex of the hexagon is -

(A) $3 + i$ (B) $(1 + \sqrt{3})i$ (C) $-3 + i$ (D) $2 + (1 - \sqrt{3})i$

Section–I(ii) : (Maximum Marks: 12)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. If $|z_1| = 1$, $|z_2 - 2| = 3$ and $|z_3 - 5| = 6$, then maximum value of $|2z_1 - 3z_2 - 4z_3|$ is not less than (where z_1, z_2, z_3 are complex numbers)-

(A) 43 (B) 44 (C) 45 (D) 46

10. Consider complex numbers z and w satisfying the equation $z+w=\frac{\bar{z}}{w}$ and $\frac{1}{w}+z=w\bar{z}$, then which of the following is/are correct ?

(A) The value of $|z|+|w|$ is 2

(B) If $\arg(z)=\pi$ then, $\arg(w)=-\frac{\pi}{3}$ or $\frac{\pi}{3}$

(C) If $w=1$, then $\text{Re}(z)=\frac{1}{2}$

(D) If $w=-1$, then $\text{Re}(z)=\frac{1}{2}$

11. Given z_1,z_2,z_3,z_4 are four points in the complex plane such that $|z_1|<1$, $|z_2|=1$ and $|z_3|\le 1$ and $z_3=\frac{z_2(z_1-z_4)}{\bar{z}_1z_4-1}$, then $|z_4|$ can be $(z_1\neq z_4)$ -

(A) 2 (B) $\frac{2}{5}$ (C) $\frac{1}{3}$ (D) $\frac{5}{2}$

Section–I(iii) : (Maximum Marks: 9)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Questions 12 - 14

Let z_1 and z_2 be two complex numbers satisfying $(z-\omega)^2+(z-\omega^2)^2=0$, where w is non real cube root of unity. Also let 'α' be a variable point on the circle $\left|z+\frac{1}{2}\right|=\frac{\sqrt{3}}{2}$ and α_1, α_2 are values of α such that $|\alpha|$ is maximum and minimum respectively.

On the basis of above information, answer the following questions :

12. Value of $|z_1-\alpha|^2+|z^2-\alpha|^2+|\omega-\alpha|^2+|\omega^2-\alpha|^2$ is -

(A) 3 (B) $\sqrt{3}$ (C) 6 (D) 1

13. Which of following is incorrect ?

(A) $|\alpha_1|+|\alpha_2|=|\alpha_1-\alpha_2|$

(B) $\alpha_1=z_2$ and $\alpha_2=z_1$ where $z_1<0$, $z_2>0$

(C) $|2z_1+1|=|2z_2+1|$

(D) $\text{amp}\left(\frac{z_1}{z_2}\right)=\text{amp}(z_1.z_2)$

14. Number of complex numbers α such that $|\alpha-\alpha_1|+|\alpha-\alpha_2|=\sqrt{6}$ is

(A) 4 (B) 3 (C) 2 (D) 0

Section–II (iii) : (Maximum Marks: 16)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. If $A=\left\{z:\left|\frac{z-2}{z+2}\right|=3,\ z\in C\right\}$ and $z_1, z_2, z_3, z_4 \in A$ are 4 complex numbers representing points P, Q, R, S respectively on the complex plane such that $z_1-z_2=z_4-z_3$, then maximum value of area of quadrilateral PQRS is

2. If α is a root of equation $x^2-x+1=0$, then the value of $|\alpha^{333}+\alpha^{666}+\alpha^{999}|$ is equal to -

3. A particle P starts from the point $z_0=1+2i$, where $i=\sqrt{-1}$. It moves first horizontally away from origin by 3 units and then vertically away from origin by 1 units to reach a point z_1. From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $-\hat{i}+\hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in clockwise direction on a circle with centre at origin, to reach a point p + iq. The value of (2p + q) is

4. Consider the curves $C_1: |z-2|=2+\text{Re}(z)$ and $C_2: |z|=3$ (where $z=x+iy$, $x,y\in R$ and $i=\sqrt{-1}$). They intersect at P and Q in the first and fourth quadrants respectively. Tangents to C_1 at P and Q intersect the x-axis at R and tangents to C_2 at P and Q intersect the x-axis at S. If area of ΔPRS is $\lambda\sqrt{2}$ sq. units, then (λ^2) is

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Section–I (i) (Maximum Marks:21)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If the domain and range of $f(x) = {}^{9-x}C_{x-1}$ contains m and n elements respectively, then -

(A) $m = n$ (B) $m = n + 1$ (C) $m = n - 1$ (D) $m = n + 2$

2. Number of ordered triplets (a, b, c) of positive integers less than 10, for which the product abc is divisible by 20, are -

(A) 48 (B) 102 (C) 54 (D) 51

3. The number of ways in which all the letters of the word 'TANATAN' can be arranged so that no two alike letters are together

(A) 24 (B) 36 (C) 38 (D) 40

4. The coefficient of x^4 in the expansion $(e^x - 1)^2$ is :

(A) $\frac{3}{8}$ (B) $\frac{5}{12}$ (C) $\frac{7}{12}$ (D) $\frac{5}{24}$

5. The sum of the series $(1^2 + 1)\,1\,! + (2^2 + 1)\,2\,! + (3^2 + 1)\,3\,! + ... + (n^2 + 1)\,n\,!$ is :

(A) $(n + 1)((n + 2)\,!)$ (B) $n\,(n + 1)\,!$ (C) $(n + 1)(n + 1)\,!$ (D) none of these

6. The value of ${}^{3n}C_n - {}^nC_1 \cdot {}^{3n-3}C_n + {}^nC_2 \cdot {}^{3n-6}C_n - {}^nC_3 \cdot {}^{3n-9}C_n + ...$ is equal to :-

(A) 1 (B) 2^n (C) 3^n (D) $3^n - 2^n$

7. The coefficient of x^4 in the expansion of $(1 + x)^{15} + (1 + x)^{16} + (1 + x)^{17} + ... + (1 + x)^{50}$ is-

(A) ${}^{51}C_5 - {}^{15}C_5$ (B) ${}^{51}C_5$ (C) ${}^{51}C_5 - 1$ (D) ${}^{52}C_6 - {}^{15}C_4$

Section–I (ii) : (Maximum Marks: 16)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. If $\sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n^2 - 4n + 6}{2 \cdot {}^nC_r}$, then -

(A) $n = 1$ (B) $n = 2$ (C) $n = 3$ (D) $n = 4$

9. If set A contains m elements, B and C are two subsets of A such that $n(B) = n(C)$; $B \cup C = A$; $n(B \cap C) = 1$ then identify the correct statements :-

(A) m must be odd

(B) m must be even

(C) Number of ways in which subset B and C are to be formed is $m \cdot {}^{m-1}C_{\frac{m-1}{2}}$

(D) Number of ways in which subsets B and C are to be formed is $\frac{1}{2} \cdot {}^{m-1}C_{\frac{m-1}{2}}$

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10. If the expansion of $(2x - 3y + z)^{10} = \sum_{\substack{0\le\alpha,\beta,\gamma\le10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!}(2x)^{\alpha}(-3y)^{\beta}(z)^{\gamma}$, then -

(A) the expansion will contain 66 terms

(B) $\sum_{\substack{0\le\alpha,\beta,\gamma\le10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} = 3^{10}$

(C) the coefficient of $x^2y^3z^5$ will be $\left(-\frac{9.10!}{5!}\right)$

(D) $\sum_{\substack{0\le\alpha,\beta,\gamma\le10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!}2^{\alpha}(-3)^{\beta} = 0$

11. The number of selections of fruits from 4 mangoes, 3 oranges, 2 papayas and 2 pineapples will be [If fruits of same type are treated as same] -

(A) 179 (when atleast one fruit is chosen)

(B) 175 (atleast 2 fruits are chosen)

(C) 15 (at most 2 fruits are chosen with atleast one fruit)

(D) 15 (none or more fruits but not more than 2 fruits are chosen)

Section–I (iii) : (Maximum Marks: 9)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Questions 12 - 14

If a man climbs either one step or k steps at a time then total number of ways in which he can climb up a staircase of 3k steps (starting from the bottom) where $k \ge 2$ is denoted as A(1, k)

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12. A(1, 2) is equal to :-

(A) 10 (B) 11 (C) 12 (D) 13

13. If A(1, k) = 26 then k is equal to :-

(A) 2 (B) 3 (C) 4 (D) k

14. If $A(1, k+1) - A(1, k) = \lambda_k$ then $\lambda_{k+1} - \lambda_k$ is equal to :-

(A) 1 (B) 2 (C) 0 (D) 4

Section–II : (Maximum Marks: 16)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. The remainder when $\sum_{r=1}^{10} \left({}^{10}C_r\right)(r-5)^3$ is divided by 10, is

2. If $n \in N$, then the remainder when $(57)^{n+3} + (79)^{n+1} + (46)n$ is divided by 11 is -

3. A gentleman gives a dinner party to 5 guests to be selected from 10 friends. The number of ways of forming a party of 5, given that two particular friends will not attend party together -

4. If the sum of all the odd divisors of 10!, which are of the form $3n + 1$ is S_1 and which are of the form $3n - 1$ is S_2. Then the remainder obtained by dividing S_1 by S_2 is

Section–I (i) (Maximum Marks:24)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. A bag contains 3 red, 2 green and 2 white balls. Two balls are drawn at random and none of them is found to be a green ball, then the chance that both are red is equal to -

(A) $\frac{1}{10}$ (B) $\frac{2}{10}$ (C) $\frac{3}{10}$ (D) $\frac{4}{10}$

2. If A & B are independent events such that $P\left(A \cap \bar{B}\right) = \frac{1}{3}$ & $P(A \cup B) = \frac{11}{15}$, then $P(A \cap B)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{5}{11}$ (C) $\frac{2}{9}$ (D) $\frac{7}{9}$

3. If 'P' is probability of an event E of a random experiment and $\{P\} = P$, then (where $\{.\}$ denotes fractional part function) -

(A) E must be a sure event

(B) E can't be a sure event

(C) $\{P(\bar{E})\} = P(\bar{E})$ for all possible values of P(E)

(D) $\{P(\bar{E})\} \neq P(\bar{E})$ for all possible values of P(E)

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4. A person has to catch a train. To catch train, from his home he can take a taxi or take rickshaw or walk by foot with respective probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$.
Probability of him catching train when he takes rickshaw from his home is half that of when the takes the taxi and probability of catching the train when he walked by foot is $\frac{1}{4}^{th}$ that of when he takes rickshaw.
He finally reached the train, the probability he walked by his foot to catch the train, is

(A) $\frac{1}{33}$ (B) $\frac{2}{33}$ (C) $\frac{7}{33}$ (D) $\frac{13}{33}$

5. A fair die is thrown repeatedly. If '3' appears on first three throws, then the probability of '4' appearing on the fourth throw, is-

(A) $\frac{1}{216}$ (B) $\frac{1}{36}$ (C) $\frac{1}{1296}$ (D) $\frac{1}{6}$

6. If $P(A) = \frac{3}{4}, P\left(A \cap B \cap \bar{C}\right) = \frac{1}{3}, P\left(A \cap \bar{B} \cap \bar{C}\right) = \frac{1}{3}$. Then $P(A \cap C) =$

(A) $\frac{1}{6}$ (B) $\frac{1}{15}$ (C) $\frac{1}{12}$ (D) $\frac{1}{9}$

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7. A biased die is such that probability of obtaining face numbered i is proportional to i. If die is rolled twice and faces 'a' and 'b' turn up on first and second turn respectively, then probability that a is even and b is odd is-

(A) $\frac{12}{49}$ (B) $\frac{9}{49}$ (C) $\frac{17}{49}$ (D) $\frac{8}{49}$

8. A bag contains 5 white and 3 black balls. 4 balls are drawn one by one, their color noted and replaced in the bag. Probability that white and black balls are drawn alternately is-

(A) $\frac{15^2}{2^{10}}$ (B) $\frac{15^2}{2^{11}}$ (C) $\frac{15^2}{2^{12}}$ (D) $\frac{15^2}{2^{13}}$

Section–I (ii) : (Maximum Marks: 12)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. In an experiment A & B events are such that $P(\bar{A}) = \frac{6}{11}; P(B) = \frac{5}{22}, P(A \cap B) = \frac{1}{11}$, then -

(A) $P(A \cup B) = \frac{15}{22}$ (B) $P(\bar{A} \cap \bar{B}) = \frac{9}{22}$

(C) $P(\bar{B} \cap A) = \frac{4}{11}$ (D) $P(\bar{A} \cup \bar{B}) = \frac{10}{11}$

10. If X and Y are independent then $P(X \cap Y) = P(X).P(Y)$. Let A & B, B & C, C & A are three pairs of independent events such that $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(C) = \frac{1}{6}$ then-

(A) $P(\bar{A} \cup \bar{B}) = \frac{5}{6}$

(B) $P(B \cap \bar{C}) = \frac{5}{18}$

(C) $P(\bar{A} \cap \bar{B}) = \frac{1}{3}$

(D) $P(\bar{A} \cap B) + P(\bar{B} \cap C) + P(\bar{C} \cap A) = \frac{25}{36}$

11. Let A, B and C be three independent events of an experiment such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(C) = \frac{1}{4}$. Which of the following statement(s) is/are correct ?

(A) Probability that atleast one of three events happening is 3/4.

(B) Probability of exactly two of these happening is $\frac{1}{4}$.

(C) Probability of atleast two of them happening is $\frac{7}{24}$.

(D) Probability of atmost two of them happening is $\frac{23}{24}$.

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Section–I (iii) : (Maximum Marks: 12)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Questions 12 and 13

Three players A, B, C alternately throw a die in the order. The first player to throw a 6 will be the winner. A's die is fair where as B and C throw a biased die with probability of throwing a 6 being p_1 and p_2 respectively.

12. Ratio of probability of winning of B to probability of winning of C is -

(A) $\frac{1-p_1}{p_1p_2}$ (B) $\frac{p_1}{1-p_1}$ (C) $\frac{p_1}{p_2(1-p_1)}$ (D) $\frac{1-p_1}{p_1}$

13. Ratio of p_1 and p_2 so that the game becomes equally likely for all the three players is -

(A) 5 : 4 (B) 1 : 2 (C) 4 : 5 (D) 2 : 1

Paragraph for Questions 14 and 15

Three sample papers x, y, z are available in the market. A student who purchase only one sample paper, purchases x, y, z with the probabilities $\frac{1}{7}, \frac{1}{3}, \frac{11}{21}$ respectively. Student will pass with the probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ if he study sample papers x, y, z respectively.

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14. Probability that student will pass in the examination is -

(A) $\frac{17}{63}$ (B) $\frac{43}{63}$ (C) $\frac{46}{63}$ (D) $\frac{23}{63}$

15. If it is known that student is fail, then probability that he studied sample paper 'y', is -

(A) $\frac{9}{92}$ (B) $\frac{55}{92}$ (C) $\frac{7}{23}$ (D) $\frac{2}{9}$

Section–II : (Maximum Marks: 16)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. If two numbers are randomly chosen from the set of first 30 natural numbers, then probability that atleast one of them is prime is given by $\frac{a}{b}$, where a & b are coprime, then value of $\frac{b-a}{19}$ will be

2. Bag I contains 4 white and 3 black balls and another bag II contains 5 white and 2 black balls. Two balls are drawn randomly from the first bag and put in the second bag and then a ball is drawn from the bag II. The probability that the ball drawn from the second bag is white is expressed as $\left(\frac{p}{q}\right)$ where p, q $\in$ N. Then the minimum value of (p + q) is

3. Humpty plays three matches against Dumpty. In any match the probabilities of Humpty getting 0, 2 and 3 points are 0.2, 0.3 and 0.5 respectively. If probability that Humpty getting atmost 6 points is $\frac{p}{200}$, then then sum of the digits of 'p' is

4. A hunter's chance of shooting an animal at a distance r is $\frac{a^2}{r^2}$ (r > a). He fires when r = 2a & if he misses he reloads & fires when r = 3a, 4a,..... If he misses at a distance '50a', the animal escapes. Find the probability that the anmial escapes.

Section–IV : (Maximum Marks: 8)

Matrix Match Type (One or More than one option correct)

For each entry in Column-I , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		Column-II
(A)	A bag contains two red, two green and a certain number of black balls. Two balls are drawn at random and the probability that they are of different colour is $\frac{5}{7}$, then number of black balls is equal to	(P)	4
(B)	Consider two independent events A & B where $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ & $P(A/\bar{B}) = \frac{a}{6}$, then 'a' is equal	(Q)	2
(C)	Minimum number of times a dice must be rolled so that the chance of getting one atleast once exceeds 0.5 is equal to	(R)	3
(D)	Odds in favour of getting exactly 3 heads in the simultaneous throw of 8 coins is "m to 25" then 'm' is equal to	(S)	odd
		(T)	prime

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Section–I(i) Single Correct Type (Maximum Marks:30)

1. If a point P moves such that its distance from line $y = \sqrt{3}x - 7$ is same as its distance from $(3\sqrt{3}, 2)$, then the area of the curve, described by P, enclosed between the coordinate axes is (in square units) -

(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{3}$

(C) $\frac{25\sqrt{3}}{2}$ (D) $25\sqrt{3}$

2. Let m_{AB} is slope of line joining the points A & B. Points $A_1(x_1, y_1)$, $A_2(x_2,y_2)$, ...& $A_n(x_n, y_n)$ are such that x_1, x_2, x_3 ..... x_n are in A.P. & y_1, y_2.....y_n are also in A.P., then $m_{A_1A_2} + m_{A_2A_3} + \ldots\ldots + m_{A_{(n-1)}A_n}$ is equal to -

(A) 0 (B) $(n-1)m_{A_1A_n}$

(C) $(n)m_{A_1A_n}$ (D) $(n^2)m_{A_1A_n}$

3. Let P(0, 3), Q(0, 5), R(x, 0), then the value of |x| for which $\angle PRQ$ is as large as possible :-

(A) $\sqrt{15}$ (B) $\sqrt{20}$

(C) $\sqrt{35}$ (D) $\sqrt{40}$

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4. The mid point of chord AB of the circle $x^2 + y^2 - 6x - 4y + 3 = 0$ is the point (1, 1). If tangents are drawn at extremities A & B, then these tangents intersect at point -

(A) (1, 0) (B) (−1, 0)

(C) (0, 1) (D) (1, −1)

5. The internal common tangents of the circles $x^2 + y^2 - 4x - 4y + 4 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are :

(A) $x - y = 2$ & $x + 2y = 3$

(B) $2x - 3y = 4$ & $x + 5 = 5$

(C) $x = 0$ & $y = 0$

(D) $x + 1 = 0$ & $y - 3 = 0$

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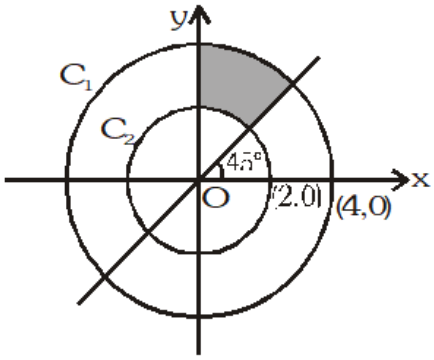
6. If the radius of the circle touching $(x-2)^2 + y^2 = 1$ and the angle bisectors of the transverse common tangents of $(x \pm 2)^2 + y^2 = 1$ is $p \pm \sqrt{q}$, then p + q is equal to -

(A) 8 (B) 9

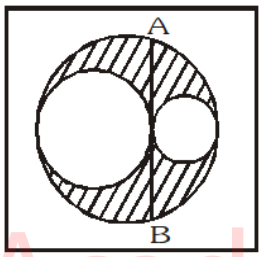
(C) 10 (D) 11

7. As shown in figure C_1 & C_2 are two concentric circles with centres at origin and respective radii are 4, 2 units. If point (a, 3) lies inside the shaded region, then interval of a will be-



(A) $a > 0$ (B) $-\sqrt{7} < a < \sqrt{7}$ (C) $a < 3$ (D) $0 < a < \sqrt{7}$

8. If the length of chord AB is 4 (as shown in adjacent figure), then find the area of shaded region -



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(A) $\frac{\pi}{2}$ (B) π (C) $\pi + 1$ (D) 2π

9. If line $y = x - 2$ is normal to a circle, co-ordinates of its centre are such that abscissa is twice of ordinate & circle touches the x-axis then equation of the circle is -

(A) $x^2 + y^2 - 8x - 4y + 16 = 0$ (B) $x^2 + y^2 - 8x - 4y + 4 = 0$

(C) $x^2 + y^2 - 4x - 2y + 2y + 4 = 0$ (D) $x^2 + y^2 - 4x - 2y + 16 = 0$

10. Tangents drawn from (6, 4) to the circle $x^2 + y^2 - 4x - 4y + 4 = 0$ intersect the y-axis at A & B. AB is equal to

(A) 6 (B) 8 (C) 10 (D) 12

Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

11. The diagonals of a square are along the pair of lines whose equation is $2x^2 - 3xy - 2y^2 = 0$. If (2, 1) is a vertex of the square, then the vertex of the square adjacent to it may be -

(A) (1, 4) (B) (−1, −4) (C) (−1, 2) (D) (1, −2)

12. ΔABC is an isosceles triangle in which A is (−1, 0), $\angle A = \frac{2\pi}{3}$, AB = AC and AB is along the x-axis. If $BC = 4\sqrt{3}$, then for the line BC -

(A) the equation $x + \sqrt{3}y = 3$ (B) x-intercept = $\sqrt{3}$(y-intercept)

(C) the equation $x - \sqrt{3}y = -3$ (D) area of triangle of $\Delta ABC = 4\sqrt{3}$

13. For the circles $S_1 = x^2 + y^2 - 4x - 6y - 12 = 0$ and $S_2 = x^2 + y^2 + 6x + 4y - 12 = 0$ and the line $L \equiv x + y = 0$

(A) L is common tangent of S_1 and S_2

(B) L is common chord of S_1 and S_2

(C) L is radical axis of S_1 and S_2

(D) L is Perpendicular to the line joining the centre of S_1 & S_2

14. C_1 & C_2 be 2 concentric circles with radii r_1 and r_2 ($r_1 < r_2$). If the tangents drawn from any point of C_2 to C_1 meet again C_2 at the ends of diameter of C_2, then -

(A) $r_2 = 2r_1$ (B) $r_2 = \sqrt{2}r_1$

(C) $r_2^2 < 2r_1^2$ (D) C_2 is director circle of C_1

Section–I(iii) Comprehension Type : (Maximum Marks: 12)

Paragraph for Questions 15 and 16

Let ABC be a triangle whose vertex A is (3, 4). $L_1 = 0$ and $L_2 = 0$ are the internal angle bisectors of angles B and C respectively, where $L_1 \equiv x + y - 2 = 0$, $L_2 \equiv 2x + y - 5 = 0$. Let A_1 is the image of point A about line L_1.

On the basis of above information answer the following :

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15. Co-ordinates of point A_1 are -

(A) (2, 1) (B) (−2, −1)

(C) (2, −1) (D) (−1, −2)

16. Inradius of triangle ABC is -

(A) $\sqrt{10}$ (B) $2\sqrt{10}$

(C) $\frac{3}{2}\sqrt{10}$ (D) $\frac{5}{2}\sqrt{10}$

Paragraph for Questions 17 and 18

Consider two circles $x^2 + y^2 = 1$ and $(x - 6)^2 + y^2 = 4$.

On the basis of above information, answer the following questions :

17. Length of direct common tangent of the circles is equal to-

(A) $\sqrt{24}$ (B) 5

(C) $\sqrt{35}$ (D) 6

18. Sum of the slopes of all the common tangents of the given circles is equal to-

(A) $3 + \sqrt{35}$ (B) 0

(C) $2\sqrt{3} + 2\sqrt{35}$ (D) $\sqrt{105}$

Section–III Integer Type : (Maximum Marks: 16)

1. A ray emitted from the point A($\cos\theta$, $\sin\theta$) strikes a mirror at a variable point P, then reflects and passes through a point B($-\sin\theta$, $\cos\theta$) such that incident and reflected rays are tangents to the circle $x^2 + y^2 = 1$, then locus of the variable point is $x^2 + y^2 = k$, then k is equal to

2. A ray of light incident at the point $(-3,-2)$ gets reflected from the tangent at $(0,-2)$ to the circle $x^2 + y^2 = 4$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is $Jx - Ky + 46 = 0$. The value of J + K is

3. The line $L_1 : 3x + 4y + 12 = 0$ is rotated through an angle of $\tan^{-1}\left(\frac{24}{7}\right)$ in anticlockwise about the point where it intersects the x-axis, line obtained in new position is denoted as $L_2 = 0$. If incentre of the triangle formed by $L_1 = 0$, $L_2 = 0$ and y-axis is $(4\lambda, 3k)$, then the value of $10k - 24\lambda$ is -

4. A line L having positive slope passing through the point (1,1) intersects the curves $2x^2 - y^2 + 2x - 4y + 2 = 0$ at points A and B. If the lines joining origin and the points A and B are such that the co-ordinate axes are the angle bisector between them then the equation of line L is given by $ax + by + 1 = 0$, then value of $(3a - b)$ is

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Section–IV Matrix Match Type : (Maximum Marks: 2)

1.

	Column-I		Column-II
(A)	If the circumference of the circle $x^2 + y^2 + 6x + 6y - b = 0$ is bisected by the circle $x^2 + y^2 - 4x + 2y + a = 0$, then $\|a + b\|$ is equal to	(P)	20
(B)	The point of intersection of the tangents of the circle $x^2 + y^2 = 10$, drawn at end points of the chord $x + y = 2$ is (a, b) then the value of $a^2 + b^2$ is equal to	(Q)	26
(C)	The intercept on the line $y = 2x$ by the circle $x^2 + y^2 - 5x = 0$ is AB. If the equation of the circle with AB as diameter is $x^2 + y^2 - ax - by = 0$ then the value of $2a + 9b$ is equal to	(R)	50
(D)	If the equation of the locus of the middle points of the chords of the circle $2x^2 + 2y^2 - 6x + 2y + 1 = 0$ that subtends an angle 120° at the centre is $x^2 + y^2 - ax - by + 2 = 0$ then $a^3 + b^3$ is equal to	(S)	42

Section–I(i) Single Correct Type (Maximum Marks:30)

1. Let $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$ are confocal ($a > A$ & $a > b$) having the foci at S_1 & S_2. Let P be their point of intersection, then S_1P & S_2P are the roots of quadratic equation -

(A) $x^2+2ax+(a^2-A^2)=0$ (B) $x^2-2Ax+(a^2+A^2)=0$

(C) $x^2-2Ax+(a^2-A^2)=0$ (D) $x^2-2ax+(a^2-A^2)=0$

2. L_1 is a tangent drawn to the curve $x^2-4y^2=16$ at A $\left(5,\frac{3}{2}\right)$. L_2 is another tangent parallel to L_1 which meets the curve at B. L_3 and L_4 are normals to the curve at A and B. Lines L_1, L_2, L_3, L_4 forms a rectangle, then which of the following is correct

(A) Equation of tangent at B is $6y=5x-16$.

(B) Equation of normal at B is $12x+10y+74=0$

(C) Radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$.

(D) Radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$

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3. From a point on the line $x-y+2=0$ tangents are drawn to the hyperbola $\frac{x^2}{6}-\frac{y^2}{2}=1$, such that the chord of contact passes through a fixed point (λ, m). Then $\frac{\lambda}{\mu}$ is equal to -

(A) 2 (B) 3 (C) 4 (D) 5

4. The points of contact Q and R of tangents from the point P(2, 3) to the parabola $y^2=4x$ are -

(A) (9, 6) and (1, 2) (B) (4, 4) and (9, 6)

(C) (9, 6) and $\left(\frac{1}{4},\ 1\right)$ (D) (1, 2) and (4, 4)

5. If $(3m_i^2, -6m_i)$ represents the feet of the normals to the parabola $y^2=12x$ from (1, 2), then $\sum\frac{1}{m_i}$ is -

(A) $-\frac{5}{2}$ (B) $\frac{3}{2}$ (C) 6 (D) -3

6. The common tangent of two parabolas $y=x^2$ and $y=x^2-2x+2$ is -

(A) $y=2x-1$ (B) $4y-x+4=0$

(C) $4x-4y-1=0$ (D) $y+4x+4=0$

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7. The locus of the point of intersection of perpendicular tangents to the parabola $x^2 - 8x + 2y + 2 = 0$ is -

(A) $2y - 15 = 0$ (B) $2y + 15 = 0$ (C) $2x + 9 = 0$ (D) none of these

8. The equation to the locus of the mid point of the line segment joining the focus of the parabola $y^2 = 8x$ to a moving point on the given parabola is -

(A) $y^2 = 4x - 1$ (B) $y^2 = 4x - 4$ (C) $x^2 = 4y - 1$ (D) $x^2 + y^2 = 1$

9. If lines $3x + 5y = 20$ and $3x - 5y = 20$ are tangents at the extremities of a latus rectum to an ellipse whose centre is origin, then length of the latus rectum is -

(A) 0.16 (B) 2.56 (C) 3.2 (D) 5.12

10. If the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ is inscribed in a square of side length $5\sqrt{2}$, then area of ellipse is -

(A) 4π (B) 8π (C) 10π (D) 12π

Section–I(ii) Multiple Correct Type : (Maximum Marks: 24)

11. Equation of common tangent(s) to $x^2 - y^2 = 12$ and $xy = 8$ is (are) -

(A) $y = 3x + 4\sqrt{6}$ (B) $y = -3x + 4\sqrt{6}$ (C) $3y = x + 4\sqrt{6}$ (D) $y = -3x - 4\sqrt{6}$

12. $y = f(x)$ is a parabola having directrix with slope 0. If $f^2(0) + f^2(1) + f^2(2) + 45 = 10f(0) + 8f(1) + 4f(2)$, then -

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(A) $y = f(x)$ is symmetric about $x = -\frac{1}{2}$ (B) $f(x) < 0 \ \forall \ x \in R$

(C) $\ell n|(f(x))|$ is one-one function (D) range of $f(x)$ is $\left(-\infty, \frac{41}{8}\right]$

13. For the parabola $x^2 + y^2 + 2xy - 4x + 4y - 4 = 0$ -

(A) The focus is (0,0) (B) The focus is $\left(-\frac{1}{2}, \frac{1}{2}\right)$

(C) The directrix is $x - y + 2 = 0$ (D) The directrix is $x - y - 2 = 0$

14. If one of the foci of an ellipse is (2, 3) whose eccentricity is $\frac{1}{3}$ and point of intersection of its major axis with the corresponding directrix is (10, 11), then identify the correct statement(s) -

(A) vertices of the ellipse are $(-2, -1)$ and (4, 5)

(B) major axis is $6\sqrt{2}$

(C) minor axis is 8

(D) latus rectum is $\frac{32\sqrt{2}}{3}$

15. Let an incident ray $L_1 = 0$ gets reflected at point A(−2, 3) on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ & passes through focus S(2, 0), then -

(A) equation of incident ray is $x + 2 = 0$
(B) equation of reflected ray is $3x + 4y = 6$
(C) eccentricity, $e = 2$
(D) length of latus rectum = 6

16. Consider four points $\left(\pm\frac{ab}{\sqrt{a^2+b^2}}, \pm\frac{ab}{\sqrt{a^2+b^2}}\right)$ which of the following is correct :-

(A) Only two ellipse can pass through them
(B) Only one ellipse can pass through them
(C) No parabola can pass through them
(D) Exactly two parabolas will pass through them

Section–I(iii) Comprehension Type : (Maximum Marks: 12)

Paragraph for Questions 17 and 18

Let minimum value of $f(x) = |x - a| + 2|x - 2|$ is 1, where $a > 2$. Consider an ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$. Tangents are drawn from point P(1, 2) to the ellipse.

On the basis of above information answer the following :

17. If 'e' is eccentricity of given ellipse, then e can be -

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{5}}{3}$ (C) $\frac{1}{3}$ (D) none of these

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18. The point of tangency of tangents from P is point Q where eccentric angle is θ, then

(A) $\sin\theta = 1$ (B) $\sec\theta = \frac{5}{3}$ (C) $\tan\theta = \frac{4}{3}$ (D) None of these

Paragraph for Questions 19 and 20

Consider three curves

$C_1 : x^2 - y^2 = 1$ $(x < 0)$

$C_2 : xy = 1$ $(y > 0)$

C_3 : Ellipse whose one focus is also focus of C_2 and corresponding directrix of this focus is asymptote of curve C_1.

Also length of major axis of C_3 is $\frac{8}{3}$.

On the basis of above information, answer the following questions :

19. Eccentricity of C_3 is equal to -

(A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

20. If A $\left(-\frac{1}{2}, -2\right)$ and BC is double ordinate of curve C_2 then locus of orthocentre of ΔABC is a -

(A) fixed point (B) parabola (C) ellipse (D) hyperbola

Section–III Integer Type : (Maximum Marks: 20)

1. If the image of the parabola $y = x^2$ in the line $x + y = 1$ is $x + y^2 = ky$, then k is equal to

2. From a variable point P, two tangents are drawn to the parabola $y^2 = 4ax$ and these tangents meet the co-ordinate axes in concyclic points. If locus of P is $x = \lambda_1 a$, then λ_1 is equal to

3. Let PQ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meeting the positive x & y axis at points P & Q respectively. Point R divides PQ internally in the ratio 2 : 1. If locus of R is $\frac{a^2}{x^2} + \frac{4b^2}{y^2} = \lambda$, then λ is equal to

4. The slope of one of the common tangents to the hyperbolas $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{y^2}{16} - \frac{x^2}{9} = 1$ is m, then |m| is

5. The chord of the hyperbola $x^2 - y^2 = 25$, touches the parabola $y^2 = 20x$. If the locus of middle point of the chords can be expressed as $x^3 = y^2 (x - a)$, then value of a is

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Section–I(i) Single Correct Type (Maximum Marks:18)

1. If latus rectum of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ subtends right angle at point (0,b), then square of eccentricity of the hyperbola is equal to -

(A) $2 - \sqrt{2}$ (B) 2 (C) $2 + \sqrt{2}$ (D) $2\sqrt{2}$

2. The slope of a chord of the parabola $y^2 = 4ax$ which is normal at one end and which subtends a right angle at the origin is-

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

3. Let a line $\ell x - y - 2\ell = 0$ (where $\ell \in R - \{0\}$) intersects the parabola $y^2 = 8x$ at points A and B. Tangents drawn to parabola at A and B intersect at C(−2, 5), then sum of the angles $\angle CAB$ and $\angle CBA$ is -

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

4. The number of values of c such that the line $y = cx + c$, $c \in R$ touches the curve $\frac{x^2}{4} + \frac{y^2}{1} = 1$ is

(A) 0 (B) 1 (C) 2 (D) more than 2

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5. Let $\frac{x}{2}\cos\theta + \frac{y}{1}\sin\theta = 1$, $\theta \in \left(\frac{\pi}{18}, \frac{\pi}{15}\right)$ intersect the ellipse $x^2 + 2y^2 = 6$ at P and Q, then angle between tangents at P and Q of the ellipse is -

(A) $\frac{\pi}{16}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

6. If mid point of the chord intercepted on the line $9x - 4y = 13$ by the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is (α, β), then $(\alpha + \beta)$ is

(A) 0 (B) 2 (C) 1 (D) −2

Section–I(ii) Multiple Correct Type : (Maximum Marks: 24)

7. Equation of a minor axis of an ellipse is $x + 2y = 3$ and length of minor axis is $\sqrt{5}$, foot of directrix is (2, 3), then

(A) equation of ellipse is $4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$

(B) eccentricity of ellipse is $\frac{1}{\sqrt{2}}$

(C) major axis is $\sqrt{10}$

(D) length of LR is $\frac{\sqrt{5}}{\sqrt{2}}$

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8. Ray $y - 2 = 0$ strike the parabola $y^2 = 4x$ at A and after reflection it passes through a point B on the axis of parabola & normal at point A meet the axis of parabola at C, then co-ordinates of -

(A) $A \equiv (2, 0)$ (B) $B \equiv (1, 0)$

(C) $B \equiv (3, -6)$ (D) $C \equiv (3, 0)$

9. From a point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, two tangents are drawn to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the corresponding chord of contact meets $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ at points Q and R then

(A) The locus of mid-point of QR is $x^2 + y^2 = a^2$

(B) The locus of mid-point of QR is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(C) The locus of image of mid-point of QR in x-axis is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(D) The locus of foot of perpendicular from point $\left(\sqrt{a^2 + b^2}, 0\right)$ to QR is $x^2 + y^2 = a^2$

10. If the eccentricity of the ellipse $\frac{x^2}{(\log a)^2} + \frac{y^2}{(\log b)^2} = 1$ $(a > b > 0, a, b \neq 1)$ is $\frac{1}{\sqrt{2}}$ and 'e' be the eccentricity of the hyperbola $\frac{x^2}{(\log_b a)^2} - y^2 = 1$, then e^2 is greater than (where $\log x = \ell n x$)-

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{4}$

11. Consider a curve C : $y = \sqrt{\frac{3x^2 - 21}{7}}$ and a line L of slope 1, which is tangent to C at point A, then

(A) Equation of L is $y = x + 2$

(B) Equation of L is $y = x - 2$

(C) Equation of circle passing through origin and touches the curve C at A is $x^2 + y^2 + x - 41y = 0$

(D) Equation of circle passing through origin and touches the curve C at A is $4x^2 + 4y^2 + x - 41y = 0$

12. If the tangent at the point P having abscissa $\frac{\sqrt{325}}{2}$ to the curve $\frac{x^2}{25} - \frac{y^2}{16} = 1$ meets the director circle of the circle $2x^2 + 2y^2 = 25$ at (α, β) & (γ, δ), then numerical value of harmonic mean of β and δ is greater than

(A) 7 (B) 6 (C) 5 (D) 4

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Section–I(iii) Comprehension Type : (Maximum Marks: 12)

Paragraph for Questions 13 and 14

Let H : $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a hyperbola such that its eccentricity and eccentricity of its conjugate hyperbola are same. Let foot of the perpendicular drawn from focus of H to its one of the tangent is $\left(\frac{1}{2}, \frac{\sqrt{15}}{2}\right)$. Tangents drawn at the ends of chord of length 2 unit of the circle C : $x^2 + y^2 = a^3 - 4b^2 + 12$, meets each other at 'P' on the transverse axis of H.

On the basis of above information, answer the following questions :

13. Number of common tangent of H & C is

(A) 2 (B) 4

(C) 6 (D) 8

14. Let E : $\frac{x^2}{16} + \frac{y^2}{\beta^2} = 1$ be the ellipse which cuts H orthogonally then β is

(A) $4\sqrt{2}$ (B) $3\sqrt{2}$

(C) $2\sqrt{2}$ (D) $\sqrt{2}$

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Paragraph for Questions 15 and 16

Let x_1, x_2, x_3, x_4 be non zero real numbers. Let A,B,C,D be distinct points on the parabola $y^2 = 4x$. Suppose that $F(x_0,y_0)$ is mid point of focal chord AB and lines BC and AE are parallel lines, where $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$, $D(x_4,y_4)$ and E(2,0).

15. The value of y_3 is-

(A) $4y_0$ (B) y_0

(C) $\frac{y_0}{2}$ (D) $2y_0$

16. If $y_1y_4 = 4$, then the tangent at D and normal at A to the parabola meet at a point whose abscissa is-

(A) $\frac{1 - x_0}{2}$ (B) $1 + x_0 - x_2$

(C) $1 - x_0 + x_2$ (D) $\frac{1 + x_0}{2}$

Section–I(v) Matching list type : (Maximum Marks: 6)

17. A hyperbola 'H' intersects the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ orthogonally. The eccentricity of the hyperbola 'H' is three times the eccentricity of the ellipse. Transverse and conjugate axis of H are along x and y axis respectively. If 2a, 2b are respectively the length of transverse axis and conjugate axis, $(x_1, 0)$ is the focus and ℓ is length of latus rectum of H.
Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	a	(1)	1
(Q)	$\|x_1\|$	(2)	2
(R)	b^2	(3)	3
(S)	ℓ	(4)	6

(A) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 4
(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 4;S $\longrightarrow$ 3
(C) P $\longrightarrow$ 4;Q $\longrightarrow$ 3;R $\longrightarrow$ 2;S $\longrightarrow$ 1
(D) P $\longrightarrow$ 1;Q $\longrightarrow$ 4;R $\longrightarrow$ 3;S $\longrightarrow$ 2

18. Let $\ell x + my + 1 = 0$ is common tangent of curves $x^2 = 4y$, $x^2 + \frac{y^2}{3} = \frac{1}{4}$ and $y^2 = 4ax$, $a > 0$.
Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	$4\ell^2 - a$	(1)	neither prime nor composite number
(Q)	$3m^2 + 3a$	(2)	odd composite number
(R)	$\frac{5\ell - a}{4m}$	(3)	odd prime number
(S)	$2a^2 + 3\ell^2 + 4m$	(4)	even composite number

(A) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 4;S $\longrightarrow$ 3
(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 4;R $\longrightarrow$ 2;S $\longrightarrow$ 3
(C) P $\longrightarrow$ 2;Q $\longrightarrow$ 3;R $\longrightarrow$ 4;S $\longrightarrow$ 1
(D) P $\longrightarrow$ 3;Q $\longrightarrow$ 4;R $\longrightarrow$ 1;S $\longrightarrow$ 2

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Section–II(iii) Numerical Type : (Maximum Marks: 16)

1. Area of triangle formed by common tangents to parabola $y^2 = 8x$ and hyperbola $\frac{x^2}{1}-\frac{y^2}{3}=1$ and directrix of parabola is

2. A parabola passes through two points A(1,0) and B(−1,0) such that directrix always touches the $x^2 + y^2 = 4$ and if locus of focus of parabola is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\frac{a^2}{b^2}$ is

3. Let A be any point on the ellipse $x^2 + 16y^2 = 4$ with foci S_1 and S_2 then the maximum value of $AS_1.AS_2$ is

4. From any point on line $(t + 2)(x + y) = 1$, $t \neq -2$ tangents are drawn to ellipse $4x^2 + 16y^2 = 1$. If chord of contact passes through a fixed point, then number of integral values of t for which fixed point lies inside ellipse is

Section–I(i) Single Correct Type (Maximum Marks:24)

1. A square matrix P satisfies $P^2 = I - P$ where I is an identity matrix. If $P^n = 5I - 8P$, then n is-

(A) 4 (B) 5 (C) 6 (D) 7

2. If maximum and minimum values of the determinant $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$ are α and β respectively then which one of the option is incorrect ?

(A) $\alpha + \beta^{99} = 4$

(B) $\alpha^3 - \beta^{17} = 26$

(C) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in N$

(D) A triangle can be constructed having it's sides as α, β and $5\alpha - 13\beta$.

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3. If a_0, a_1, a_2, a_3, a_4 are in A.P. with commom difference d (where $d \neq 0$ & ± 1) , then the value of $\begin{vmatrix} a_1a_2 & a_1 & a_0 \\ a_2a_3 & a_2 & a_1 \\ a_3a_4 & a_3 & a_2 \end{vmatrix}$ is

(A) $2d^4$ (B) $2d^3$ (C) $2d^2$ (D) $4d^4$

4. Which of the following statement is correct about two square matrices A and B of same order -

(A) trace (Adj(AB)) = Adj(trace (AB).I)

(B) trace ((A + B) (A − B)) ≠ trace (A^2) − trace(B^2)

(C) trace(Adj(|A| |B| AB)) − trace(Adj(|AB| BA)) = 0

(D) If A is symmetric matrix and B is skew symmetric matrix then trace (AB' − BA') ≠ 0

5. If A is a skew symmetric matrix of order 2 and B, C are $\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$, $\begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix}$ respectively, then $A(B^2C^2)AT$ is -

(A) a symmetric matrix (B) an identity matrix

(C) a skew symmetric matrix (D) neither symmetric nor skew symmetric

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6. Let B be a square matrix such that $|B| = 1$ and $2A + B = B^T - A^T$, then $|A + B|$ is -

(A) 0 (B) -1 (C) 1 (D) 2

7. If A is a square matrix of order 'n' with $|A| = 9$ and $|adjA| = 3^{16}$, then number of elements in A, is -

(A) 81 (B) 27 (C) 9 (D) 3

8. Number of solutions of equation $\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \cos\left(\theta + \frac{\pi}{6}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & -\sin\left(2\theta + \frac{\pi}{3}\right) \\ -\cos\left(\theta - \frac{\pi}{6}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{2\pi}{3}\right) \end{vmatrix} = 0$ in $[0,2\pi]$ is-

(A) 2 (B) 4 (C) 8 (D) infinite

Section–I(ii) Multiple Correct Type : (Maximum Marks: 8)

9. System of equations $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

(A) unique solution when $a \neq 0, b \in R$

(B) no solution when $a = 0, b = 1$

(C) infinite solutions when $a = 0, b = 2$

(D) infinite solutions when $a = 1, b \in R$

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10. If $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ and $A^3 - 2A^2 + \lambda A + I = O$ (Null matrix), then-

(A) $\lambda = 4$ (B) $\lambda = -4$

(C) $|A^2 + A^{-1}| = 56$ (D) $|A^2 + A^{-1}| = 14$

Section–I(iii) Comprehension Type : (Maximum Marks: 15)

Paragraph for Questions 11 and 12

If A is invertible matrix of order 3 and B is another matrix of same order as of A such that $|B| = 2$ and $A^T|A|B = A|B|B^T$, then
(where adj(A) and A^{-1} denote adjoint and inverse of matrix A)

11. $\left|AB^{-1}\text{adj}\left(A^TB\right)^{-1}\right|$ is equal to-

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{16}$ (D) 1

12. If B is symmetric matrix , then which of the following is incorrect ?

(A) adj(B) is symmetric matrix

(B) B^{-1} is symmetric matrix

(C) $A^{2015}\,B^{2016}$ is symmetric matrix

(D) AB^{-1} is symmetric matrix if AB = BA

Paragraph for Questions 13 - 15

$$g(y) = \begin{vmatrix} p-y & q & r \\ r & p-y & q \\ q & r & p-y \end{vmatrix}, \quad p > q > r > 0$$

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On the basis of above information answer the following :

13. Absolute term in g(y) can be -

(A) 2 (B) -1 (C) 0 (D) all of the above

14. Coefficient of y in g(y) will be -

(A) independent of only p

(B) independent of q as well as r

(C) independent of only r

(D) none of these

15. If y_1, y_2 & y_3 are the roots of $y^3 = 1$ such that $f(y) = p + qy + ry^2$, then g(0) is equal to -

(A) $f(y_1) \cdot f(y_2) \cdot f(y_3)$

(B) $f(y_1) + f(y_2) + f(y_3)$

(C) $\frac{f(y_1) + f(y_2) + f(y_3)}{3}$

(D) $\{f(y_1) \cdot f(y_2) \cdot f(y_3)\}^{1/3}$

Section–III Numerical Type : (Maximum Marks: 16)

1. If system of equations

$x + (\sin\alpha)y + (\sin^2\alpha)z = 0,$

$x + (\cos\alpha)y + (\cos^2\alpha)z = 0$

$x + (\sin 2\alpha)y + (\sin^2 2\alpha)z = 0$

has non trivial solutions, then number of distinct values of α (where $\alpha \in [0,\pi]$), is

2. Let $a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 = \begin{vmatrix} x^2+x & (x-2) & 1 \\ (2x+3) & x & 3 \\ (x-5) & -1 & (x+1) \end{vmatrix}$ is an identity w.r.t x, then the value of $(a_4 + a_3 + a_2 + a_1 + a_0) - 25$ is

3. Let $A_1 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \neq 0$ & $A_2 = \begin{vmatrix} a+2b & b+3c & c+4a \\ b+2c & c+3a & a+4b \\ c+2a & a+3b & b+4c \end{vmatrix}$, then $\frac{A_2}{A_1}$ is -

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4. The number of all possible symmetric matrices of order 3×3 with entries $-1,0,1$ and whose trace equals 1, is

Section–I(i) Single Correct Type (Maximum Marks:21)

1. A polynomial function satisfies $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$, $x \neq 0$ and $f(3) = -26$, then value of $f(4)$ is-

(A) -15 (B) -63 (C) -47 (D) -255

2. Let $f(x)$ be polynomial of degree 5 with leading coefficient unity such that $f(1) = 5$, $f(2) = 4$, $f(3) = 3$, $f(4) = 2$, $f(5) = 1$, then $f(6)$ is equal to -

(A) 0 (B) 24 (C) 120 (D) 720

3. A and B are two sets such that n(A) = 4 and n(B) = 5. If number of possible mappings defined from set A to B is x and out of these y mappings are one-one, then x + y will be -

(A) 720 (B) 740 (C) 745 (D) 900

4. Range of the function $f(x) = \frac{2|x| - 1}{3|x| + 2}$ is-

(A) $\left(-\frac{2}{3}, \frac{2}{3}\right)$ (B) $\left[-\frac{1}{2}, \frac{2}{3}\right)$ (C) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (D) $R - \left\{\frac{2}{3}\right\}$

5. Domain of the function $f(x) = \frac{1}{x} + \frac{1}{\sqrt{x^2 - 4x}} + \log x$ is-

(A) $[4,\infty)$ (B) $(4,\infty)$ (C) $[0,4]$ (D) $(-\infty,4)$

6. If $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$, $n \in N$, then the maximum value of n is-

(A) 6 (B) 4 (C) 5 (D) 30

7. The value of $\sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) + \tan^{-1}\left(2 - \sqrt{3}\right)$ is-

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{12}$

Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

8. Which of the following function(s) is/are odd function(s) ?

(A) $f(x) = \frac{x^3}{2^{|x|} + \sec x}$

(B) $f(x) = x^4 e^{\left(\sqrt{x^2+1}+x^4\right)}$

(C) $f(x) = \frac{(e^x - 1)}{(e^x + 1)}(g(x) + g(-x))$

(D) $f(x) = h(g(k(x)))$ when h(x) , g(x) & k(x) are all odd function

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9. If $f(x) = \sqrt{|x|^2 - 5|x| + 6} + \sqrt{8 + 2|x| - |x|^2}$, then x can be-

(A) $[-4,-3]$ (B) $[-3,-2]$ (C) $[-2,2]$ (D) $[3,4]$

10. $2\cot^{-1}7 + \cos^{-1}\frac{3}{5}$ is equal to-

(A) $\cot^{-1}\frac{44}{117}$ (B) $\text{cosec}^{-1}\frac{125}{117}$ (C) $\tan^{-1}\frac{4}{117}$ (D) $\cos^{-1}\frac{44}{125}$

11. The value of $\cos(2\cos^{-1}x + \sin^{-1}x)$ at $x = \frac{1}{5}$ is -

(A) negative (B) positive (C) rational (D) irrational

Section–I(iii) Comprehension Type : (Maximum Marks: 9)

Paragraph for Questions 12 - 14

Consider three functions $f(x) = x^2 - 4x$; $g(x) = x + 5$, $h(x) = x - 12$.

On the basis of above information, answer the following questions :

12. Number of integral values, which are not in the range of $\frac{g(f(x))}{h(f(x))}$ is -

(A) one (B) two (C) three (D) four

13. If the largest interval of x for which $g(f(x))$ is invertible is taken as $[a, \infty)$ and least value of $h(f(x))$ is 'b' then the value of $\sin^{-1}\left(\frac{8}{b}\right) + \cos^{-1}\left(-\frac{a^3}{b}\right) + \tan^{-1}\left(-\frac{32}{ab}\right)$ is -

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(A) $\frac{2\pi}{3}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) π

14. Value of $\lim\limits_{x\to 7}\lim\limits_{n\to\infty}\frac{[g(h(x))]^{2n} + 1}{h(f(x))}$ is equal to -

(A) $\frac{4}{9}$ (B) $\frac{2}{9}$

(C) $\frac{1}{9}$ (D) Limit does not exist

Section–III Integer Type : (Maximum Marks: 16)

1. Number of integer(s) in the range of $f(x) = \text{sgn}(\ell n(x^2 - 2x + 2))$ is (where sgn(x) represents signum function)

2. Let $f(x) = \frac{1}{(1 + \cot x)}$, then the value of $f(x) + f\left(\frac{\pi}{2} - x\right) + f\left(\frac{\pi}{2} + x\right) + f(\pi - x)$ is

3. Number of solutions of equation $\sin^{-1}(\sin 6x) = -\frac{x}{2}$ in $[0, \pi]$ is

4. Number of integral value(s) of x in the domain of $f(x) = \sin^{-1}(x - 1) + \log_{\pi}(2 - x)$ is

Section–I(i) Single Correct Type (Maximum Marks:21)

1. The value of $\tan\left(\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1+2r}{1+r^2(r+1)^2}\right)\right)$ is -

(A) 1 (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) none of these

2. If $f(x)=\frac{1}{x^2-17x+66}$, then $f\left(\frac{2}{x-2}\right)$ is discontinuous at

(A) $x=2,\ \frac{7}{3},\ \frac{25}{11}$ (B) $x=2,\ \frac{8}{3},\ \frac{24}{11}$

(C) $x=2,\ \frac{7}{3},\ \frac{24}{11}$ (D) no where in its domain

3. If $f(x)=\cot^{-1}\left(\frac{2x}{1-x^2}\right)$ and $g(x)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ then the value of $\lim\limits_{x\to t}\frac{f(x)-f(t)}{g(x)-g(t)}$, $(0<t<1)$ is -

(A) 1 (B) -1

(C) 2 (D) 1/2

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4. Let $f(x)$ is a non-zero differentiable function $\forall\ x\in R$ & satisfying $f(x+y)=f(x-y)\ \forall\ x, y\in R$, then -

(A) $f(1)+f(0)=f'(0)$ (B) $f(1)-f(0)=f'(0)$

(C) $f(0)+f(1)=f(2)$ (D) $3f(2)=2f(3)$

5. If $f(x)=\sqrt{x+\sqrt{x-\sqrt{x+\sqrt{x-\sqrt{x}.....\infty}}}}$, then value of $\lim\limits_{x\to1^+}\frac{f(x)-\left(\frac{\sqrt{x}+1}{2}\right)}{(x-1)}$

(A) 1 (B) 0 (C) $\frac{3}{8}$ (D) $\frac{3}{4}$

6. Let $f(x)=\begin{cases}a+(x-b)^2 & \text{for } |x-b|\le k\\ c+|x-b| & \text{for } |x-b|>k\end{cases}$, then the positive value of k so that f(x) becomes differentiable at $x=b-k$ and $x=b+k$ is-

(A) $\frac{3}{2}$ (B) 1/2 (C) 1 (D) 2

7. If $f(x)=\lim\limits_{n\to\infty}\frac{e^x+e^{-x}x^{2n}}{e^xx^{2n}+e^{-x}}$, then -

(A) $f(1^+)=e^2$ (B) $f(1^-)=\frac{1}{e^2}$ (C) $f(1)=e^2$ (D) $f(1^+)=\frac{1}{e^2}$

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Section–I(ii) Multiple Correct Type : (Maximum Marks: 12)

8. Let a,b,c be positive numbers such that a + b + c = 15. Then the value of $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{1/x}$ can be -

(A) 5 (B) 4 (C) $\frac{3}{2}$ (D) 6

9. Let [x] be the greatest integer function $f(x) = \frac{\sin\frac{1}{4}\pi[x]}{[x]}$ is -

(A) not continuous at any point

(B) continuous at $\frac{3}{2}$

(C) discontinuous at 2

(D) differentiable at $\frac{4}{3}$

10. If $f(x) = \begin{cases} x^m \cos\left(x-\frac{1}{x}\right) & x>0 \\ 0 & x=0 \end{cases}$, then -

(A) $\forall\ m \in (-\infty, 0]$, $f(x)$ is discontinuous at x = 0

(B) $\forall\ m \in (0, \infty)$, $f(x)$ is differentiable at x = 0

(C) $\forall\ m \in (1,\infty)$, $f(x)$ is differentiable at x = 0

(D) $\forall\ m \in (0, 1]$, $f(x)$ is continuous but not differentiable at x = 0

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Section–I(iii) Comprehension Type : (Maximum Marks: 6)

Paragraph for Questions 11 and 12

Consider the following functions

$f(x) = \begin{cases} 1-x, & 1 \leqslant x \leqslant 2 \\ 3-x, & 2 < x \leqslant 3 \end{cases}$

$g(x) = x^4 + x^2 + 1$

$h(x) = x^3$

On the basis of above informations, answer the following questions :

11. The function $gof(x)$ is -

(A) discontinuous at $x = 2$

(B) continuous but not derivable at $x = 2$

(C) continuous and derivable at $x = 2$

(D) non derivable at more that one point in $[1, 3]$

12. The function $h \circ f(x)$ is -

(A) discontinuous at $x = 2$

(B) continuous but not derivable at $x = 2$

(C) continuous and derivable at $x = 2$

(D) increasing in $(1, 2)$

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Section–III Integer Type : (Maximum Marks: 12)

1. The function $\max.\{1-x, 1+x, 2x^2\}$; $x \in R$ is discontinuous at 'p' points and non-differentiable at q points then $p + q$ is

2. If $f : R \longrightarrow R^+$ is differentiable function such that $|f(y).\, f(x-y)| + (x-y)^2 \geq |f(x)\, f(x-y)| \forall\ x, y \in R$, $f(0) = 5$, then the value of $f(5)$ is

3. If $f(x) = \begin{cases} \tan^{-1}\left(\frac{ax+b}{c}\right) & 0 < |x| < \frac{1}{2} \\ 0 & x = \frac{1}{2} \\ \log\left(2+bx^2\right) & \frac{1}{2} < x < \frac{1}{\sqrt{2}} \end{cases}$ is differentiable in $x \in \left(0, \frac{1}{\sqrt{2}}\right)$, then $\left|\frac{a+b}{c}\right|$ is equal to

Section–IV Matrix Match Type : (Maximum Marks: 6)

1.

	Column-I		Column-II
(A)	The number of the values of x in $(0, 2\pi)$, where the function $f(x) = \frac{\tan x + \cot x}{2} - \left\lvert \frac{\tan x - \cot x}{2} \right\rvert$ is continuous but not differentiable is	(P)	2
(B)	The number of points where the function $f(x) = \min\{1, 1 + x^3, x^2 - 3x + 3\}$ is non-derivable	(Q)	0
(C)	The number of points where $f(x) = (x + 4)^{1/3}$ is non-differentiable is	(R)	4
(D)	Consider $f(x) = \begin{cases} -\frac{\pi}{2}\ell n\left(\frac{x.2}{\pi}\right) + \frac{\pi}{2}, & 0 < x \leqslant \frac{\pi}{2} \\ \sin^{-1}\sin x, & \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$ Number of points in $\left(0, \frac{3\pi}{2}\right)$, where $f(x)$ is non-differentiable is	(S)	1

2.

	Column - I		Column - II
(A)	Number of points where the function $f(x) = \begin{cases} 1 + \left[\cos\frac{\pi x}{2}\right] & 1 < x \leqslant 2 \\ 1 - \{x\} & 0 \leqslant x < 1 \\ \lvert\sin \pi x\rvert & -1 \leqslant x < 0 \end{cases}$ and $f(1) = 0$ is continuous but non-differentiable where [ ] denote greatest integer and { } denote fractional part function	(P)	0
(B)	$f(x) = \begin{cases} x^2 e^{1/x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f'(0^-)$ is equal to	(Q)	1
(C)	The number of points at which $g(x) = \frac{1}{1 + \frac{2}{f(x)}}$ is not differentiable where $f(x) = \frac{1}{1 + \frac{1}{x}}$, is	(R)	2
(D)	Number of points where tangent does not exist for the curve $y = \text{sgn}(x^2 - 1)$	(S)	3

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Section–I(i) Single Correct Type (Maximum Marks:30)

1. Let g is the inverse function of differentiable function f & $f'(x) = \frac{x^{10}}{1+x^2}$. If g (2) = a then g'(2) is equal to -

(A) $\frac{5}{2^{10}}$ (B) $\frac{1+a^2}{a^{10}}$ (C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$

2. If $y = \tan\theta + \cot\theta$, $x = \tan\theta - \cot\theta$, where θ is a parameter then $y^3\frac{d^2y}{dx^2}$ is equal to-

(A) 2 (B) -2 (C) 4 (D) -4

3. If $f(x) = \sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{(x-1)}}$, then $f'(x)$ at x = 1.5 is -

(A) 0 (B) $-\sqrt{2}$ (C) $-\sqrt{3}$ (D) -4

4. If the normal to the curve $y = f(x)$ at $x = 0$ be given by $3x - y + 3 = 0$, then the value of $\lim\limits_{x\to1}(x-1)^2\left\{f((x-1)^2) - 5f(4(x-1)^2) + 4f(7(x-1)^2)\right\}^{-1}$ is -

(A) $-\frac{1}{3}$ (B) $\frac{1}{3}$ (C) 3 (D) -3

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5. In order that the function $f(x) = (x+1)^{\cot x}$ be continuous at x = 0, then f(0) must be equal to :

(A) 0 (B) $\frac{1}{e}$ (C) 1 (D) e

6. $\lim\limits_{x\to1}(x-1)\ell n(\sin^{-1}(\ell nx))$ is equal to :-

(A) 0 (B) $\frac{\pi}{2}$ (C) $\ell n\frac{\pi}{2}$ (D) 1

7. Let $f(x) = \begin{cases} g(x).\cos\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ where g(x) is an even function differentiable at x = 0, passing through the origin. Then f '(0)

(A) is equal to 1 (B) is equal to 0

(C) is equal to 2 (D) does not exist

8. $\lim\limits_{x\to\infty}\frac{\cos^{-1}\left(\sqrt{x+\sqrt{x+\sqrt{x}}} - \sqrt{x+\sqrt{x}}\right)}{\sec^{-1}\left(-x^2\ell n\left(\frac{x^2+2}{x^2+1}\right)\right)}$ is equal to -

(A) 1 (B) $\frac{\pi}{2}$ (C) $\frac{1}{2}$ (D) π

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9. Let $g : R \to R$ be a differentiable function such that, $\lim\limits_{\alpha \to \frac{\pi}{2}} g(2009 + \cos\alpha) = 3$ and $\lim\limits_{n\to\infty} n\left(g\left(2009 + \frac{1}{n}\right) - g(2009)\right) = 27$, then $\lim\limits_{t\to 0^+}\left(\frac{g(2009)}{g(2009+t)}\right)^{1/t}$ is :-

(A) e^3 (B) e^2

(C) e^{-2} (D) e^{-9}

10. Let $f(u) = \frac{1}{\ell n|u| - 1}$, $u = \frac{x+e}{x+1}$. Number of values of x in domain for which $f(u)$ is discontinous is -

(A) 2 (B) 1

(C) 3 (D) 0

Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

11. If g is inverse of f and $f(x) = x^2 + 3x - 3$ $(x > 0)$ then g'(1) equals -

(A) $\frac{1}{2g(1)+3}$ (B) -1

(C) $\frac{1}{5}$ (D) $\frac{-f'(1)}{(f(1))^2}$

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12. If $\lim\limits_{x\to\infty}\left(\frac{ax+2}{\sqrt{4+x^2}}\right)^{\frac{ax^3+1}{x^2+1}} = L$, where $a \geq 0$ & L is finite number, then -

(A) $a = 0, L = 1$ (B) $a = 1, L = e^2$

(C) $0 < a < 1, L = 0$ (D) $0 \leq a < 1, L = 0$

13. $f(x) = (\sin^{-1}x)^2 . \cos(1/x)$ if $x \neq 0$, $f(0) = 0$, f(x) is :

(A) continuous no where in $-1 \leqslant x \leqslant 1$

(B) continuous every where in $-1 \leqslant x \leqslant 1, x \neq 0$

(C) differentiable no where in $-1 \leqslant x \leqslant 1$

(D) differentiable everywhere $-1 < x < 1, x \neq 0$

14. If $f(x) = \cos\left[\frac{\pi}{x}\right]\cos\left(\frac{\pi}{2}(x-1)\right)$; where [x] is the greatest integer function of x, then f(x) is continuous at-

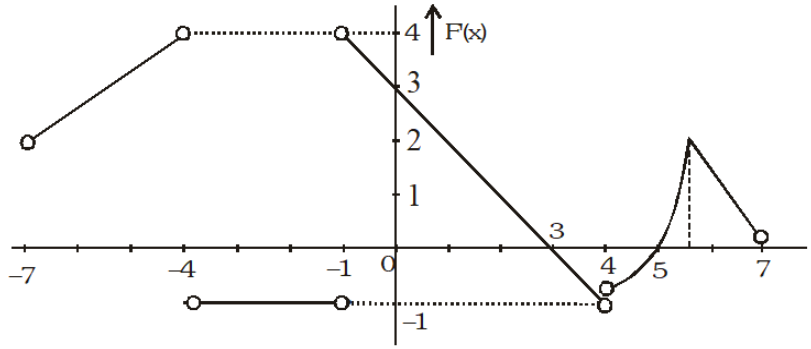
(A) $x = 0$ (B) $x = 1$

(C) $x = 2$ (D) none of these

Section–I(iii) Comprehension Type : (Maximum Marks: 6)

Paragraph for Questions 15 and 16

Graph of the derivative of a continuous function F(x) in (−7, 7) is given below



On the basis of above information answer the following :

15. F(x) is a linear function in :-

(A) $(-7, -4)$ (B) $(-4, -1)$ (C) $(-1, 4)$ (D) $\left(\frac{11}{2}, 7\right)$

16. F'(x) does not exist at :-

(A) 1 points (B) 2 points

(C) 3 points (D) 4 points

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Section–III Integer Type : (Maximum Marks: 16)

1. If $f : R \longrightarrow R$ is a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(4)$, then find $f(1)$

2. Let $y(x) = \sin(x - \sin x)$ and $y''(x)$ denotes the second order derivative of $y(x)$ w.r.t.x. The value of $\left[y''\left(\frac{\pi}{2}\right) + \cos 1\right]$ is- (where [.] denotes the greatest integer function).

3. If function $f(x) = \begin{cases} \frac{\sqrt[3]{7+x^3}-\sqrt{3+x^2}}{(x-1)} & , \ x \neq 1 \\ b & , \ x = 1 \end{cases}$ (where $a \in N$ and $b \in R$) is continuous at x = 1, then value of a + 4b is

4. Number of points at which $f(x) = \sin\left([x]\frac{\pi}{2}\right) \operatorname{sgn}(x^2 - 3x + 2)$ is discontinuous in $x \in [0, 3]$, is (where [.] and sgn (.) denotes greatest integer and signum function respectively)

Section–IV Matrix Match Type : (Maximum Marks: 2)

1. Consider a function $f(x)$ in $[0, 2\pi]$

defined as $f(x) = \begin{cases} [\sin x] + [\cos x], & 0 \leqslant x \leqslant \pi \\ [\sin x] - [\cos x], & \pi < x \leqslant 2\pi \end{cases}$

where [.] denotes greatest integer function.

On the basis of above match the entries of column-I with the corresponding entries in column-II.

	Column-I		Column-II
(A)	Number of points where $f(x)$ is discontinuous is equal to	(P)	2
(B)	Number of points, where $\lim_{x\to a} f(x)$ does not exist is equal to (where $a \in (0, 2\pi)$)	(Q)	3
(C)	Number of points where $f(x)$ is non derivable is equal to	(R)	4
(D)	Minimum value of $f(x)$ is m then $\lvert m \rvert$ is	(S)	prime
		(T)	neither composite nor prime

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Section–I(i) Single Correct Type (Maximum Marks:21)

1. If f (x) is an even function which is also periodic with the period T and $\int_0^a f(x)dx = 3$ and $\int_{-T/2}^{3T/2} f(x)dx = 18$, then $\int_{-a}^{a+5T} f(x)dx$ is equal to -

(A) 96 (B) 93 (C) 51 (D) 48

2. The value of $\int_{3/2}^{15} [\ell n[x]]dx$ is equal to, (where [.] denotes greatest integer function) -

(A) $\frac{19}{2}$ (B) 9 (C) 19 (D) $\frac{9}{2}$

3. Let for every integer a, $\int_a^{a+1} f(x)dx = a^2$, then the value of $\int_{-2}^{4} f(x)dx$ is equal to -

(A) 16 (B) 17 (C) 18 (D) 19

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4. The value of definite integral $\int_{-1}^{1} \frac{x\tan^{-1}x}{1+e^{\tan^{-1}x}}dx$ is :-

(A) $\frac{\pi}{4} - \ell n2$ (B) $\frac{\pi}{4} - \frac{1}{2}$ (C) $\frac{\pi}{4} + \frac{1}{2}$ (D) $\frac{\pi}{2} - 1$

5. $\int \frac{x^2-4x+6}{x^2-2x+4}dx$ is equal to -

(A) $x + \ell n(x^2 - 2x + 4) + c$ (B) $\ell n(x^2 - 2x + 4) + 2x + c$

(C) $2x - \ell n(x^2 - 2x + 4) + c$ (D) $x - \ell n(x^2 - 2x + 4) + c$

6. $\int \frac{(1-\tan^2 x).\text{cosec}\quad 2xdx}{\sec^2 x}$ is equal to

(A) $\frac{\ell n|\sin 2x|}{2} + C$ (B) $\frac{\ell n|\cos 2x|}{2} + C$

(C) $\frac{\ell n(\tan x)}{2} + C$ (D) $\frac{\ell n|\cos x + \sin x|}{2} + C$

7. If $\int e^x(\ell nx + x\ell nx + 1)dx = f(x) + C$ when $f(1) = 0$, then $f(e)$ is equal to-

(A) e (B) e^e (C) e^{e-1} (D) e^{e+1}

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Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

8. If $I_{(n)} = \int_0^{\pi} (1 - |\sin x| + |\sin 2x| - |\sin 3x| + \ldots\ldots\ldots \text{ upto n terms})\, dx$ then -

(A) $I_{(2)} = \pi - 2$

(B) $I_{(3)} = \pi$

(C) $I_{(2008)} = \pi - 2$

(D) $I_{(2009)} = \pi$

9. If $I_n = \int_0^1 (1 - x^2)^n dx$, then -

(A) $I_n = \frac{2n}{2n+1} I_{n-1}$

(B) $I_n = \frac{2.4.6.....2n}{3.5.7.....(2n+1)}$

(C) $I_n = \frac{2^n n!}{3.5.7.....(2n+1)}$

(D) $I_n = \frac{2n+1}{2n} I_{n-1}$

10. $\int \frac{3e^x + 5e^{-x}}{4e^x - 5e^{-x}} dx = Ax + B\,\ell n|4e^{2x} - 5| + C$ then -

(A) A is -1

(B) B is $\frac{7}{8}$

(C) C can be -1

(D) $A = 1$ & $B = -\frac{7}{8}$

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11. Let $f : [0,\infty) \longrightarrow R$ be a continuous strictly increasing function, such that $f^3(x) = \int_0^x tf^2(t)dt \ \forall\ x \geq 0$, then which of the following is/are **INCORRECT**-

(A) $f(x)$ is onto function

(B) $\int_0^1 f(x)dx = \frac{1}{18}$

(C) Number solutions of $6f(x) = e^x$ is 2

(D) graph of $y = 6f(x)$ intersects the graph of $y = 3x^2 + 2x + 2$ at one point

Section–III Integer Type : (Maximum Marks: 20)

1. Let a differentiable function $f(x)$ satisfies $f(x)\,f'(-x) = f(-x)\,f'(x)$ and $f(0) = 1$, then $\int_{-2}^{2} \frac{dx}{(1+f(x))}$ is equal to

2. The value of integral $\int_{1/2}^{2} e^{\left|x-\frac{1}{x}\right|}dx$ is equal to $\lambda_1\left[e^{\lambda_2/2}-1\right]$ then $\lambda_1+\lambda_2$ is equal to

3. If $I_1 = \int_{0}^{100\pi} (\sin(\cos x)+\cos(\sin x))dx$ & $I_2 = \int_{0}^{1} \frac{\cos x dx}{\sqrt{1-x^2}}$, then $\frac{I_1}{50I_2}$ is equal to

4. If $\lim\limits_{n\to\infty} \sum\limits_{r=n}^{2n} \frac{n^4}{r^3\sqrt{r^4+n^4}} = \sqrt{a}-\sqrt{b}$, then $\frac{32b}{a}-10$ is equal to

5. $\int \left(x^{12}+x^8+x^4\right)\left(2x^8+3x^4+6\right)^{1/4}dx = a\left(2x^{12}+3x^8+6x^4\right)^{5/4}+C$ (where C is integration constant), then the value of $(60a+1)$ is

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Section–IV Matrix Match Type : (Maximum Marks: 4)

1.

	Column-I		Column-II
(A)	$\int_4^{10} \frac{[x^2]dx}{[x^2-28x+196]+[x^2]}$ = {where [.] denotes greatest integer function}	(P)	$\frac{1}{100}$
(B)	$\int_{-1}^{2} \frac{\|x\|}{x}dx =$	(Q)	3
(C)	$\lim_{n\to\infty} \frac{1^{99}+2^{99}+.....+n^{99}}{n^{100}} =$	(R)	$\frac{1}{3}$
(D)	$5050\int_{-1}^{1}\sqrt{x^{200}}dx = \frac{1}{\alpha}$, then α =	(S)	1

2.

	Column-I		Column-II
(A)	Let $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, then n + 2m is equal to	(P)	2
(B)	Let $\int \frac{xdx}{x^2-8x+15} = \ell n\left\|\frac{(x-5)^a}{(x-3)^b}\right\| + c$, then a + b is equal to	(Q)	3
(C)	Let y = f(t) & x = g(t) where t is a parameter & f,g are twice derivable functions. If $f'(1) = 0$, g'(1)=1 & $f''(1)=2$ then $\frac{d^2y}{dx^2}$ at t = 1 is equal to	(R)	0
(D)	Let L = $\lim_{x\to 0^+} (-\ell nx)^{\frac{1}{\ell nx}}$, then ℓnL is equal to	(S)	−2
		(T)	Natural number

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Section–I(i) Single Correct Type (Maximum Marks:24)

1. $\int \log(\sin^2 x.\cos x).\dfrac{(2-3\sin^2 x)}{\sin 2x}dx$ equals -
(where C is the constant of integration)

(A) $\dfrac{1}{4}\log^2\left(\dfrac{1}{2}\sin 2x.\sin x\right)+C$ (B) $\log\left(\dfrac{1}{2}\sin 2x.\sin x\right)+C$

(C) $\log^2(\sin 2x.\sin x)+C$ (D) $\log(\sin 2x.\sin x)+C$

2. If $\int \ell n\left(\sqrt{2+x}+\sqrt{2-x}\right)dx = x\,\ell n\left(\sqrt{2+x}+\sqrt{2-x}\right)+A\sin^{-1}\dfrac{x}{2}+Bx+C$, then $|B|$ is equal to
(where C is constant of integration)

(A) $\dfrac{1}{2}$ (B) 1 (C) 2 (D) 3

3. If $f(x)=\max\left\{x^2,(1-x)^2,\dfrac{3}{4}\right\}$, $x\in[0,1]$, then the value of $\int_0^1 f(x)\,dx$ is

(A) $\dfrac{6\sqrt{3}-1}{12}$ (B) $\dfrac{3}{4}(\sqrt{3}-1)$ (C) $\dfrac{2-\sqrt{3}}{2}$ (D) $\dfrac{8-3\sqrt{3}}{24}$

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4. $\int_{\pi/3}^{\pi/6} \ell n\sin x\,dx-\int_{-\ell n2}^{\frac{1}{2}\ell n\frac{3}{4}} \sin^{-1}e^x dx$ is equal to

(A) $\dfrac{\pi}{6}\ell n\dfrac{2}{3}$ (B) $\dfrac{\pi}{6}\ell n\dfrac{3}{2}$ (C) $\dfrac{\pi}{3}\ell n\dfrac{3}{4}$ (D) $\pi\,\ell n\dfrac{4}{3}$

5. Given $\int_0^y e^t dt+\int_0^x f(t)dt=g(x)$, then $\dfrac{dy}{dx}$ is equal to (where $f(x)$ is continuous and $g(x)$ is a differentiable function)

(A) $\dfrac{g'(x)-f(x)}{1+g(x)-\int_0^x f(t)dt}$ (B) $\dfrac{g'(x)-f(x)}{g(x)-\int_0^x f(t)dt}$ (C) $e^y(g'(x)-f(x))$ (D) $\dfrac{(g'(x)-f(x)).e^y}{g(x)-\int_0^x f(t)dt}$

6. $\int e^x\left(\dfrac{1}{\ell nx}-\dfrac{1}{(\ell nx)(\ell nx^x)}\right)dx$ is equal to-
(where C is constant of integration)

(A) $\dfrac{e^x}{\ell nx}+C$ (B) $e^x\ell nx+C$ (C) $2e^x.\log_x e+C$ (D) $\dfrac{e^x}{(\ell nx)^2}+C$

7. If $f(x)$ is satisfies $\lim\limits_{x\to 0} \dfrac{x\,\ell n(1+x)}{\int\limits_0^x f(t)\sin(t)\,dt} = 5$ and continuous, then $f(0)$ is equal to-

(A) $\dfrac{2}{5}$ (B) 10 (C) 5 (D) $\dfrac{5}{2}$

8. If $I_1 = \int\limits_0^1 \dfrac{e^x}{1+x}dx$, $I_2 = \int\limits_0^1 \dfrac{x^2}{e^{x^3}(2-x^3)}dx$, then $\dfrac{I_1}{I_2}$ is equal to

(A) 3e (B) $\dfrac{3}{e}$ (C) $\dfrac{1}{3e}$ (D) $\dfrac{e}{3}$

Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

9. If $\int \sqrt{\sec^2 x + 3}\,dx = \ell n\left|f(x) + \sqrt{4+f^2(x)}\right| + \sqrt{3}\sin^{-1}\left(\dfrac{k.\,g(x)}{2}\right) + C$, where C is an integration constant, then

(A) value of k is equal to $\sqrt{3}$ (B) $g(x) = \dfrac{f(x)}{\sqrt{1+f^2(x)}}$

(C) $g^2(x) + \left(\sqrt{1+f^2(x)}\right)^{-2} = 1$ (D) $g^2(x)f^2(x) = f^2(x) - g^2(x)$

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10. If $\int \left(2x^6 + 15x^4 + 2x^2 + 3\right)\cos(2x)dx = f(x)\sin(2x) + g(x)\cos(2x) + K$, where $f(x)$ and $g(x)$ are polynomial of x such that $f(0) = 1$, $g(-1) = -4$ and K is constant of integration, then

(A) $4f(1) \le 3g(1)$ (B) $\int\limits_{-\frac{\pi}{3}}^{\frac{\pi}{3}} f(x)dx \ne 0$ (C) $g''(0) < 10$ (D) $\dfrac{g(2)}{f(2)} > 1$

11. Let $f(x-3) = f(x+3)$ for all $x \in R$ and $f(x) = \begin{cases} x & ; \ 0 \le x < 3 \\ 6-x & ; \ 3 \le x < 6 \end{cases}$, then

(A) $\int\limits_0^{36} f(x)dx = 54$ (B) $\int\limits_0^{48} f(x)dx = 64$ (C) $\int\limits_0^{54} f(x)dx = 81$ (D) $\int\limits_0^{18} f(x)dx = 9$

12. Let $f(x)$ is differentiable bijective function such that $f'(x) - \int\limits_0^x f(t)\,dt = \int\limits_1^{f(x)} g(t)\,dt\ \forall x > 0, f(0) = 1$ and $g(x) = f^{-1}(x)$, then-

(A) $g(e) = \sqrt{2}$ (B) $g(1) = \sqrt{e}$ (C) $f(1) = \sqrt{e}$ (D) $f(e) = \sqrt{2}$

Section–I(iii) Matching list type : (Maximum Marks: 3)

13. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	$\lim\limits_{n\to\infty}\left[\int\limits_0^2\left(1+\frac{t}{n+1}\right)^n dt\right]$ equals to	(1)	$e-\frac{1}{2}e^2-\frac{3}{2}$
(Q)	Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and g be the function satisfying $f(x) + g(x) = x^2$, then value of $\int\limits_0^1 f(x)\,g(x)\,dx$ is	(2)	e^2
(R)	$\int\limits_0^1 e^{(e^x)}(1+xe^x)\,dx$ is equals to	(3)	e^2-1
(S)	$\lim\limits_{k\to0}\frac{1}{k}\int\limits_0^k(1+\sin 2x)^{\frac{1}{x}}dx$, is equals to	(4)	e^e

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(A) P ⟶ 4;Q ⟶ 2;R ⟶ 3;S ⟶ 1

(B) P ⟶ 3;Q ⟶ 1;R ⟶ 4;S ⟶ 2

(C) P ⟶ 4;Q ⟶ 1;R ⟶ 2;S ⟶ 3

(D) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4

Section–III Integer Type : (Maximum Marks: 24)

1. If $f(x)$ is an even function and $\alpha = \int_{2017}^{2019} (x-2017)^2 f(x-2017)\,dx$ and $\beta = \int_{-2019}^{-2015} (x+2017)^2 f(x+2017)\,dx$, then $\frac{\beta}{\alpha}$ is $(\alpha \neq 0)$

2. The value of integral $\int_0^4 \min\{|x-1|, |x-2|, |x-3|\}\,dx = \frac{a}{b}$ where a and b are co-prime, then the value of $(a + b)$ is

3. If $\int_0^{10} f(x)\,dx = 9$, then value of $\sum_{r=1}^{10} \int_0^1 f(x+r-1)\,dx$ is

4. Let $f : R \longrightarrow R$ be non-constant differentiable function and satisfies, $f(x) = x^2 - \int_0^1 (x + f(t))^2 dt$. Then $f(4)$ is equal to

5. $\frac{1}{n\pi} \int_{-n}^{n} \frac{dx}{\sqrt{\{x\}\{-x\}}}$, $n \in N$ is equal to (where $\{.\}$ fractional part function)

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6. Value of $\frac{29\int_0^1 (1-x^4)^7 dx}{4\int_0^1 (1-x^4)^6 dx}$ is equal to

Section–I(i) Single Correct Type (Maximum Marks:21)

1. Let $f(x) = x^3 + ax + b$ with $a \neq b$ and suppose the tangent lines to the graph of f at $x = a$ and $x = b$ have the same gradient. Then the value of $f(1)$ is equal to -

(A) 0 (B) 1 (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

2. If the points of local extremum of $f(x) = x^3 - 3ax^2 + 3(a^2 - 1)x + 1$ lies between -2 & 4, then 'a' belongs to -

(A) $(-2,2)$ (B) $(-\infty, -1) \cup (3, \infty)$

(C) $(-1, 3)$ (D) $(3, \infty)$

3. Let f be a function which is continuous and differentiable for all real x. If $f(2) = -4$ and $f'(x) \geq 6$ for all $x \in [2, 4]$ then

(A) $f(4) < 8$ (B) $f(4) \geq 8$ (C) $f(4) \geq 12$ (D) none of these

4. If x and y are real numbers such that $x^2 + y^2 = 27$, then maximum possible value of $x - y$ is -

(A) $\sqrt{72}$ (B) $\sqrt{18}$ (C) $\sqrt{54}$ (D) $\sqrt{\frac{2}{3}}$

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5. A boat leaves the dock at 2 PM and travels due south at a speed of 20 km/h. Another boat has been heading due east at 15 km/hr and reaches the same dock at 3 pm. At what time the two boats were closest to each other -

(A) 2:21:36 PM (B) 2:15:00 PM

(C) 2:15:15 PM (D) 2:21:40 PM

6. Consider the function for $x = [-2, 3]$, $f(x) = \begin{cases} \frac{x^3-2x^2-5x+6}{x-1} & \text{if } x \neq 1 \\ -6 & \text{if } x = 1 \end{cases}$, then

(A) f is discontinuous at $x = 1 \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$

(B) $f(-2) \neq f(3) \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$

(C) f is not derivable in $(-2, 3) \Rightarrow$ Rolle's theorem is not applicable

(D) Rolle's theorem is applicable as f satisfies all the conditions and c of Rolle's theorem is 1/2

7. If $x.y = 1$, then minimum value of $x^2 + y^2$ is -

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) 4

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Section–I(ii) Multiple Correct Type : (Maximum Marks: 12)

8. If $f(x) = 3x^2 + \int_0^x e^{-t} f(x-t)dt$, then

(A) $f(x) = 0$ has 3 solutions

(B) $f(x)$ is monotonic increasing

(C) $f(x) = 4$ has 2 solutions

(D) $\int_{-1}^{1} f(x)dx = 2$

9. Let f be a differentiable function $\forall\ x \in R$ where $f(1) = 1$ & $f(3) = 4$ & $f'(x) \geq 1\ \forall\ x \in R$, then $f(2)$ can be equal to

(A) 1 (B) 2

(C) 3 (D) 4

10. Let $f(x)$ is a derivable function which is increasing for all $x \in R$ (having no critical point), then

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(A) $f(3 - 4x)$ is an increasing function for all x.

(B) $f(3 - 4x)$ is a decreasing function for all x.

(C) $f(x^2 - x)$ increases for $x > \frac{1}{2}$.

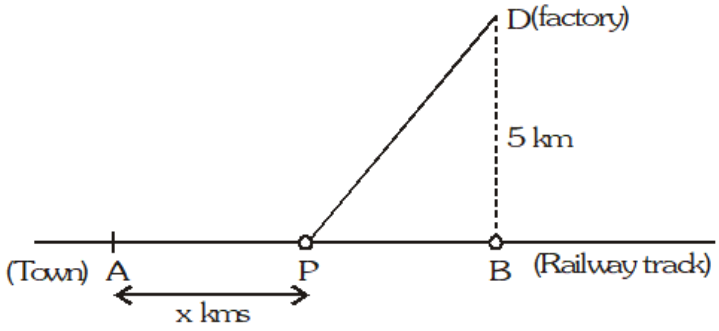
(D) $(f(x))^3$ is an increasing function for all x.

Section–I(iii) Comprehension Type : (Maximum Marks: 12)

Paragraph for Questions 11 and 12

A factory is connected with a straight railway track by a highway. Town A lies on the railway track. The distance DB from the factory to railway track is 5 km. The segment AB of railway track is equal to 'ℓ' kms. Freight charges on highway are 2 times higher than on railway. A highway DP is connected to railway track such that AP is x kms (see the figure).

On the basis of above information answer the following :



11. If 'ℓ' is $\frac{185}{\sqrt{3}}$ kms, then the value of 'x' for which freight charge is minimum :-

(A) $\frac{60}{\sqrt{3}}$ kms (B) $60\sqrt{3}$ kms (C) $120\sqrt{3}$ kms (D) $75\sqrt{3}$ kms

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12. If ℓ = 10 kms, then the value of x for which freight charges are maximum :-

(A) 10 kms (B) $10 - \frac{5}{\sqrt{3}}$ kms (C) $\frac{10}{\sqrt{3}}$ kms (D) 0 km

Paragraph for Questions 13 and 14

If $f : R \longrightarrow R$ is a function defined by $f(x) = 2\cos^2 2x + \frac{3}{4}\sin 4x + ax$, where $a \in R$.

On the basis of above information, answer the following :

13. Complete set of values of 'a' for which $f(x)$ is strictly increasing for all $x \in R$, is -

(A) $(-\infty, -5]$ (B) $[5, \infty)$ (C) $(-5, \infty)$ (D) $(-\infty, 5]$

14. Complete set of values of 'a' for which $f(x)$ does not have any critical point, is -

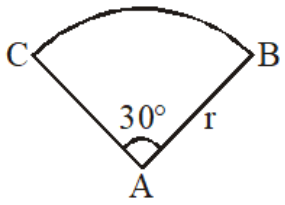
(A) $(-\infty, -5]$ (B) $[5, \infty)$

(C) $(-5, 5)$ (D) $(-\infty, -5) \cup (5, \infty)$

Section–III Integer Type : (Maximum Marks: 16)

1. If m is the slope of a line which is tangent to $y^3 = x^4$ & a normal to $x^2 - 2x + y^2 = 0$, then $\left(\frac{3m}{4}\right)^3$ is equal to $(m \neq 0)$

2. Consider a circular arc ABC as shown in figure, having variable radius r. The circle passing through points A, B, C is having radius R. The value of $\left(2\cos 15^\circ \times \frac{dR}{dr}\right)$ is equal to



3. Water is flowing in a container at a rate proportional to the square of the amount of water present inside the container. If at t = 0, volume of water inside the tank is V_0 & it takes 6 hours for the volume to become $2V_0$, then the time (in hours) taken for the volume to become $4V_0$ is equal to

4. Let $f(x)$ be a function defined by $f(x) = \int_3^x t\left(t^2 - 7t + 12\right) dt$, $3 \leq x \leq 6$. If the range of $f(x)$ is [a, b], then the value of $\left(3a + \frac{b}{9}\right)$ is

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Section–I(i) Single Correct Type (Maximum Marks:30)

1. The set of values of 'a' for which the function $f(x) = (a+2)x^3 - 3ax^2 + 9ax - 1$ is decreasing for all real x, is -

(A) $a \le -3$ (B) $a \le -2$ (C) $a \ge -3$ (D) $a \ge 3$

2. Complete set of values of m, for which function $f(x) = 3 + mx + e^{-x}$ is always decreasing, is -

(A) $[0, \infty)$ (B) $(-\infty, 0]$ (C) $[2, 5]$ (D) $[7, 17]$

3. Which of the following curve has constant length of subnormal at every point -

(A) $x^2 = y$ (B) $y^2 = x$ (C) $x^2 + y^2 = 1$ (D) $x^2 - y^2 = 1$

4. Let f, g are two continuous and twice derivable functions such that $f(0) = f(3) = 0$; $f(1).f(2) < 0$; $g(0) = g(3) = 0$. Number of roots of equation $f''(x).g(x) + f'(x).g'(x) = 0$ in $(-1,5)$ cannot be

(A) 2 (B) 3 (C) 4 (D) 5

5. Given $f(x) = \begin{cases} |x-1|+5 & \text{if } x \ne 1 \\ 8 & \text{if } x = 1 \end{cases}$, then at $x = 1$, $f(x)$ has-

(A) a local maximum (B) a local minimum

(C) neither maximum nor minimum (D) a unique tangent with finite slope.

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6. Function $f(x) = \dfrac{x^2-2}{\sqrt{1+x^2}}$

(A) is always increasing (B) is always decreasing

(C) has exactly one point of minima (D) has exactly one point of maxima

7. If $f(x) = \begin{cases} 2x+3 & x < 0 \\ 5 & x = 0 \\ x^2+7 & x > 0 \end{cases}$, then at $x = 0$, $f(x)$

(A) is decreasing (B) is increasing

(C) is neither increasing nor decreasing (D) has local maxima

8. Let $f(x) = \displaystyle\int_0^{x^2} (t-1)(t-4)(t-9)dt$, then

(A) $f''(x) = 0$ have 4 distinct positive solutions (B) $f'''(x) = 0$ have 2 distinct positive solutions

(C) $f'''(x) = 0$ have 3 distinct positive solutions (D) $f(x)$ have 6 critical points.

9. Number of non negative integers which satisfy the inequation $f(f(f(x))) > f(f(-x))$, where $f(x) = 30 - 2x - x^3$ is-

(A) 0 (B) 1 (C) 2 (D) 3

10. A box is to be made with square base and open top. If the area of material used is 48 sq. meter, then maximum volume of the box is-

(A) $48m^3$ (B) $16m^3$ (C) $32m^3$ (D) $36m^3$

Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

11. Let the parabola $y = x^2 + ax + b$ and $y = x(c - x)$ touches each other at point (1,0), then-

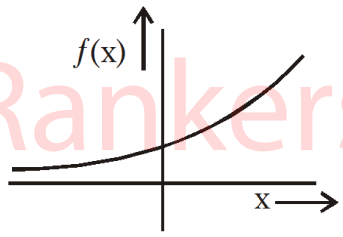
(A) $a + c = -2$ (B) $b + c = 3$ (C) $a = -3$ (D) $b = 2$

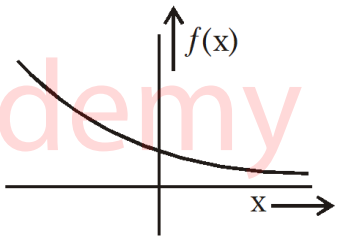
12. Let $f(x) = \begin{cases} (x-2)^2 + 3 & x \geq 2 \\ k + x^2 & x < 2 \end{cases}$, then

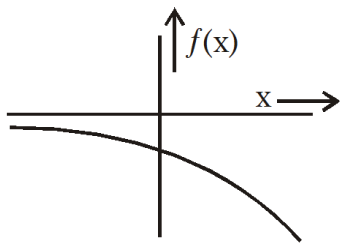
(A) $f(x)$ have local minima at $x = 2$ $\forall$ $k \geq -1$ (B) Range of $f(x)$ is $[3, \infty)$ $\forall$ $k \in [3, \infty)$

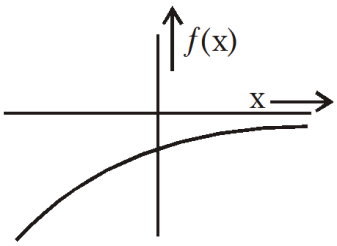
(C) $f(x)$ is non-differentiable $\forall$ $k \in R$ (D) $f(x)$ is continuous $\forall$ $k \in R$

13. Let $f(x)f'(x)f''(x) > 0$ $\forall$ $x \in R$, then which of the following is(are) best possible graph of $f(x)$

(A) 

(B) 

(C) 

(D) 

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14. Let a function $f : R \longrightarrow R$ satisfies functional equation $f(x + y) = f(x).f(y)$ where $f(0) \neq 0$ and $f'(0) = 1$. Given $g(x) = \dfrac{3x}{f(x)}$, then -

(A) The value of $g'(\ell nx)$ at $x = \sqrt{e}$ is $\dfrac{3}{2}(e)^{-\frac{1}{2}}$

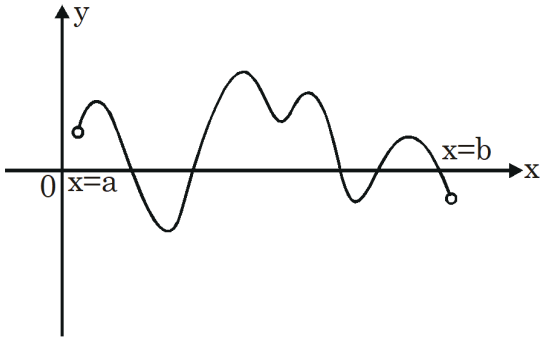
(B) g(x) attain local minima at x = 1

(C) g(x) attain local maxima at x = 0

(D) if g(x) = k has two solutions then $k \in \left(0, \dfrac{3}{e}\right)$

Section–I(v) : Matching list type : (Maximum Marks: 3)

15. Consider the curve for y = $f'(x)$ as shown below



then for the curve y = $f(x)$.

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Number of points of maxima	(1)	7
(Q)	Number of points of minima	(2)	3
(R)	Number of inflection points	(3)	5
(S)	Number of stationary points	(4)	2

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(A) P ⟶ 3;Q ⟶ 4;R ⟶ 1;S ⟶ 2

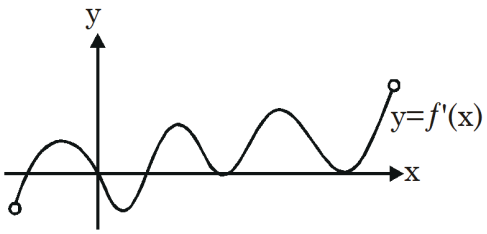
(B) P ⟶ 4;Q ⟶ 1;R ⟶ 2;S ⟶ 3

(C) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 3

(D) P ⟶ 2;Q ⟶ 4;R ⟶ 1;S ⟶ 3

Section–(II): (iii) Integer Type : (Maximum Marks: 20)

1. Let $f : R \longrightarrow R$ and graph of function $y = f'(x)$ is shown below



Let a = number of point of inflection of $y = f(x)$

b = number of point of local minima of $f(x)$

c = number of point of local maxima of $f(x)$

d = number of critical point where neither maxima nor minima occur, then value $\frac{a - c}{a + b + d}$ is equal to

2. Let $f(x) = \frac{e^x}{x}$. If $f(x) = k$ has two distinct solution then smallest integral value of k is

3. Let $f(x) = \begin{cases} e^x - 2x, & 1 \leqslant x \leqslant 2 \\ 2\sin x + x, & 2 < x \leqslant 2\pi \end{cases}$, then least value of $f(x)$ in $x \in [1, 2\pi]$ is equal to

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4. The number of distinct real roots of equation $2x^4 - 8x^3 + 8x^2 - 1 = 0$ is n, then the value of $\frac{1}{n}$ is

5. If tangent to the curve $xy = x^2 + 1$ at (α,β) is normal to the curve $x^2 + y^2 + 2fy + c = 0$, then $\frac{1}{|f.\alpha|}$ is

Section–I(i) Single Correct Type (Maximum Marks:24)

1. Area bounded by $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and $y = 3x - 15$, for $x > 1$, is -

(A) 73 (B) 13/6 (C) 73/6 (D) none of these

2. A tangent line with positive slope to a curve $y = x^2 + 1$ is drawn such that area bounded by curve, tangent line and co-ordinate axes is $\frac{11}{3}$ then x co-ordinate of point of tangency is

(A) 5 (B) 3 (C) 7 (D) 1

3. Consider a function $f(x) = \int_0^x [\cos t]\,dt$, $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ then area bounded by curve $y = f(x)$, x axis and lines $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ is (where [.] denotes greatest integer function) :-

(A) π (B) π^2 (C) $\frac{\pi^2}{2} - 1$ (D) $\frac{\pi^2}{2}$

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4. The area of the smaller portion above x-axis bounded by the curves $y^2 = 8x$ and $\frac{(x-2)^2}{4} + \frac{y^2}{16} = 1$, is -

(A) $4\pi - \frac{32}{3}$ (B) $8\pi - \frac{16}{3}$ (C) $8\pi - \frac{64}{3}$ (D) $2\pi - \frac{16}{3}$

5. The general solution of the differential equation $\frac{dy}{dx} = \left(x^3 - 2x\tan^{-1}y\right)\left(1 + y^2\right)$ is -

(A) $2\tan^{-1}x = y^2 - 1 + 2ce^{-x^2}$ (B) $2\tan^{-1}y = x^2 - 1 + 2ce^{-x^2}$

(C) $2\tan^{-1}y = y^2 - 1 + 2ce^{-x^2}$ (D) $2\tan^{-1}x = x^2 - 1 + 2ce^{-y^2}$

6. If $y = e^{(K+1)x}$ is a solution of differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$, then k =

(A) -1 (B) 0 (C) 1 (D) 2

7. The order and degree of the differential equation $\sqrt[3]{\frac{dy}{dx}} - 4\frac{d^2y}{dx^2} - 7x = 0$ are a and b, then a + b is -

(A) 3 (B) 4 (C) 5 (D) 6

8. The area bounded by the x-axis, part of the curve $y = 1 + x^{-2}$ and the ordinates $x = 1$ & $x = 2$ is divided into equal parts by the ordinate at $x = a$ such that $a = \frac{3 + \sqrt{p}}{8}$, then the value of p is -

(A) 70 (B) 71 (C) 72 (D) 73

Section–I(ii) Multiple Correct Type : (Maximum Marks: 20)

9. The general solution of the differential equation $e^x dx = \sqrt{1-e^{2x}}dy$ is -

(A) $y = K - \cos^{-1}(e^x)$ (B) $y = \sin^{-1}(e^x) + C$

(C) $y = \sin^{-1}\sqrt{1-e^{2x}} + C$ (D) $y = \cos^{-1}\sqrt{1-e^{2x}} + C$

10. Solution of the differential equation $b(x^2 + a^2)dy + a(y^2 + b^2)dx = 0$ is

(A) $bx + ay = c(ab - xy)$ (B) $\tan^{-1}\left(\frac{x}{a}\right) + \tan^{-1}\left(\frac{y}{b}\right) = c$

(C) $\cot^{-1}\left(\frac{a}{x}\right) - \tan^{-1}\left(\frac{y}{b}\right) = c$ (D) $\tan^{-1}\left(\frac{x}{a}\right) - \cot^{-1}\left(\frac{b}{y}\right) = c$

11. A function $y = f(x)$ satisfies the differential equation $\frac{d}{dx}(xy) = 1$ and $f(1) = 2$, then :-

(A) f periodic function (B) f is one-one function

(C) domain is of f is $R - \{0\}$ (D) function is neither odd nor even

12. If A denotes the area between the curve $2\{y\} = [x] + 1$ for $0 \le y < 1$ and the x-axis between $x^2 - x \le 0$, then A is less than or equal to (where {.} and [.] are the fractional part and greatest integer functions respectively) -

(A) $\frac{\pi}{6}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

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13. Area enclosed by the curve $y = \sin x$ with x-axis between $x = 2n\pi$ to $x = 2(n+1)\pi$, is-

(A) $\int_0^{2\pi} \sin x\,dx$ (B) $2\int_0^{\pi} \sin x\,dx$ (C) $4\int_0^{\pi/2} \sin x\,dx$ (D) 4

Section–II Integer Type : (Maximum Marks: 20)

1. If the area bounded by x-axis, y-axis, x = 2, y=2 & $xy \le 2 \le 8(x^2 + y^2)$ is $\alpha - \frac{\pi}{\beta} + \ell n\gamma$ where $\alpha,\beta,\gamma \in N$ then $\beta - 2(\alpha + \gamma)$ is equal to

2. Area bounded by the straight lines $x^2y - y^3 - x^2 + 5y^2 - 8y + 4 = 0$ is equal to (in sq. units)

3. If $\frac{dy}{dx} = \frac{e^x(\sin x + \cos x)}{(\tan y + y\sec^2 y)}$ and the curve $y = f(x)$ passes through origin & (x_1, π), where $6 < x_1 < 7$, then $\frac{x_1}{\pi}$ is equal to

4. If $y = ax^p + bx^{-q}$ where $p,q \in N$ & a,b are constants such that $x^2y'' + 2xy' = 12y$ then 'p+q' is -

5. If $(2xy - y^2 - y)dx = (2xy + x - x^2)dy$ and $y(1) = 1$, then the value of $12|y(-1)|$ is

MATHS

(Quadratic Equation and Sequence Series) GUIDED REVISION # 18

TIME : 60 Min. M.M. : 78

Section–I(i) Single Correct Type (Maximum Marks:24)

1. If α,β are the integral roots of the quadratic equation $x^2 - \alpha\beta x + \alpha^2 + \beta = 0$, then number of possible different quadratic equation(s) is-

(A) 0 (B) 1

(C) 2 (D) 4

2. If roots of the quadratic equation $(x-\alpha)(x-\beta)+(x-\gamma)(x-\delta)=0$ are m & n respectively, then roots of the equation $2(x-m)(x-n)-(x-\alpha)(x-\beta)=0$ will be

(A) $\frac{\alpha+m+\gamma}{2}, \frac{\beta+n+\delta}{2}$ (B) γ, δ

(C) $\alpha-m, \beta-n$ (D) $m-\alpha+\gamma, n-\beta+\delta$

3. If $a \in (0, 10)$ such that roots of the equation $x^2 - 4x - a^2 + 2a + 3 = 0$ lies between the roots of the equation $x^2 - 4x + 4 - a^2 = 0$ then number of integral values of 'a' are -

(A) 5 (B) 6

(C) 8 (D) 9

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4. If α & β are real roots of $(\log_3 x)^2 - 6\log_3 x + k = 0$, $k \in R$, then minimum integral value of $(\alpha+\beta)$ is-

(A) 53 (B) 54

(C) 55 (D) 56

5. Let $a_1, a_2, \ldots\ldots a_n$ be a sequence such that $a_1 = 2$ and $a_n - a_{n-1} = n^2 \ \forall\ n \geq 2$. If $a_k = 205$ then k is equal to -

(A) 5 (B) 8

(C) 11 (D) 15

6. If common ratio of a geometric progression is positive then maximum value of the ratio of second term to the sum of first 3 terms equals -

(A) 2 (B) $\frac{1}{3}$ (C) 3 (D) $\frac{1}{2}$

7. The minimum value of $\frac{12}{x} + \frac{18}{y} + xy$, where x and y are positive real numbers, is -

(A) $12\sqrt{3}$ (B) $24\sqrt{3}$

(C) 18 (D) infinity

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8. Let A_i, i = 1, 2,....6 represents 6 AMs
G_i, i = 1,2,3,.....6 represents 6 GMs
H_i, i = 1, 2, ...,6 represents 6 HMs between two numbers 2 & 72
and G represents single GM between 2 & 72, then the value of $\frac{\left(\prod_{i=1}^{6} G_i\right)\left(\sum_{i=1}^{6} \frac{1}{H_i}\right)}{\left(\sum_{i=1}^{6} A_i\right)}$ is -

(A) G (B) G^2

(C) G^4 (D) G^6

Section–I(ii) Multiple Correct Type : (Maximum Marks: 16)

9. If equations $ax^2 - (a + b)x + b = 0$ & $bx^2 + (b - c)x - c = 0$ has exactly one root in common {a, b, c ≠ 0}, then which of the following can be correct -

(A) $b^2 = ac$ (B) $-a = b \neq c$

(C) $b = a \neq c$ (D) $-a \neq b = c$

10. Let $\alpha, \beta, \gamma, \delta$ be the roots of $x^4 - x^3 - x^2 - 1 = 0$. Also consider $p(x) = x^6 - x^5 - x^3 - x^2 - x$. Then the value of $p(\alpha) + p(\beta) + p(\gamma) + p(\delta)$ cannot be

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(A) 4 (B) 5

(C) 6 (D) 7

11. Two consecutive numbers k, k +1 are removed from natural numbers 1,2,3,.....n such that arithmetic mean reduces by 1, then (n − k) is equal to -

(A) Prime number

(B) Neither prime nor composite number

(C) Odd number

(D) Even number

12. If positive numbers x, y, z are in A.P., then the minimum value of $\frac{x+y}{2y-x} + \frac{y+z}{2y-z}$ is less than -

(A) 2 (B) 6

(C) 8 (D) −1

Section-I(iii) Comprehension Type : (Maximum Marks: 6)

Paragraph for Question 13 to 14

Let $f(x) = 2x^2 - 2(2\lambda + 1)x + \lambda(\lambda + 1) = 0$, where λ is a parameter.

On the basis of above information, answer the following questions :

13. If both roots of equation $f(x) = 0$ belong to $(-1,1)$, then greatest integral value of $|10\lambda|$ is -

(A) 2 (B) 4 (C) 5 (D) 9

14. If equation $ax^2 + bx + c = 0$, where a, b & c are sides of any triangle and equation $f(x) = 0$ having both roots common, then set of possible values of λ is -

(A) $(-\infty, -2)$ (B) $(-5, 0)$ (C) $(-5, -4)$ (D) $(4, \infty)$

Section–II (iii) : Integer Type : (Maximum Marks: 24)

1. Let α, β are the roots of the equation $x^2 + ax + b = 0$ and γ,δ are the roots of $x^2 - ax + b - 2 = 0$. If $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{5}{12}$ & $\alpha\beta\gamma\delta = 24$, then the value of 'a' is

2. The only integer which does not lie in the range of the function $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$ is -

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3. If $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...$ upto n terms is equal to 1575, then the value of n will be

4. A number consists of three digits which are in G.P. The sum of the right hand and left hand digit exceeds twice the middle digit by 1 and the sum of the left hand and middle digits is two third of the sum of the middle and right hand digit. Find the numbers.

5. $1 + \frac{(1+2)(1+3)}{2^2(1^3+2^3)} + \frac{(1+2+3)(1+3+5)}{3^2(1^3+2^3+3^3)} + \frac{(1+2+3+4)(1+3+5+7)}{4^2(1^3+2^3+3^3+4^3)} + ...$ upto10 terms $= \frac{a}{b}$ (where a & b are relatively prime), then $a - b$ is equal to

6. If $(1 + 3 + 5 + ... + a) + (1 + 3 + 5 + ... + b) = (1 + 3 + 5 ..... + c)$, such that (i) $a + b + c = 57$ and (ii) $a < 10$, then the value of $(2b - 2c + a)$ is equal to

Section–IV : Matrix Match Type : (Maximum Marks: 8)

Matrix Match Type (One or More than one option correct)

For each entry in Column-I, +2 If only the bubble(s) corresponding to all the correct match(es) is(are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		Column-II
(A)	If p^{th}, q^{th} and r^{th} terms of a G.P. (common ratio $\neq 1$) are in G.P. then p, q, r will be in	(P)	Arithmetic progression
(B)	If a,b,c,d are non-zero real numbers such that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) < (ab + bc + cd)^2$ and $c(a - b) = a(b - c)$, then a,b,c,d will be in	(Q)	Geometric progression
(C)	If a,b,c,d are four different positive numbers in G.P. then $\frac{a(2\log a + \log b)}{b}, \frac{b(2\log b + \log c)}{c}, \frac{c(2\log c + \log d)}{d}$ will be in	(R)	Harmonic progression
(D)	If a_1,a_2,a_3 are in A.P., a_2,a_3,a_4 are in G.P. and a_3,a_4,a_5 are in H.P. then a_1,a_3,a_5 are always in	(S)	Not a harmonic progression
		(T)	A progression, whose all the terms are identical

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Section–I(i) Single Correct Type (Maximum Marks:30)

1. $\tan200^\circ(\cot10^\circ - \tan10^\circ) + 1$ is equal to -

(A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) 1 (D) 3

2. If $\frac{(\cos1^\circ+\sin1^\circ)(\cos2^\circ+\sin2^\circ)(\cos3^\circ+\sin3^\circ)....(\cos45^\circ+\sin45^\circ)}{\cos1^\circ.\cos2^\circ\cos3^\circ.....\cos45^\circ} = a^b$, where a & b are prime numbers and a < b, then a + b is equal to -

(A) 22 (B) 23 (C) 24 (D) 25

3. If $\sin\frac{\pi}{8} = \sqrt{\frac{1}{\sqrt{m}} - \frac{1}{\sqrt{n}}}$ where m, n ∈ N then m+n is equal to-

(A) 8 (B) 6 (C) 12 (D) 10

4. A right angled trapezium is circimscribed about a circle. The radius of the circle, if the lengths of the bases (i.e. parallel sides) are equal to a and b is :

(A) a + b (B) $\frac{ab}{a+b}$ (C) $\frac{a+b}{ab}$ (D) |a - 1-b|

5. If $\log_\alpha 8 = \gamma$, $\log_\beta a = -1$ and $\log_{1/4}\beta = -1$ then $\left(\frac{1}{\alpha}+1\right)^{\log_{\sqrt{5}}(\beta^2+4\gamma^2)}$ is equal to -

(A) $\sqrt{5}$ (B) 5 (C) 25 (D) 625

6. Number of solution(s) of the equation $\log_2(x^2+3) = \frac{1}{2}\log_{1/3}\left(x+\frac{1}{x}\right)$, x > 0 is -

(A) 0 (B) 1 (C) 2 (D) infinite

7. For θ = 100°, the value of sinθ + cosθ is -

(A) less than 0 (B) between 0 and 1

(C) equal to 0 (D) between 1 and $\sqrt{2}$

8. Sum of the solutions of the equation $\sin^2\theta\tan\theta + \cos^2\theta\cot\theta - \sin2\theta = 1 + \tan\theta + \cot\theta$, where $0 \leqslant \theta \leqslant \pi$, is -

(A) 3π (B) $\frac{3\pi}{2}$ (C) $\frac{5\pi}{2}$ (D) 7π

9. If $2^{\sec^2x}\sqrt{\tan^2y - \tan y + \frac{1}{2}} \leqslant 1$, then number of values of x can be, x ∈ [0, 2π] -

(A) 0 (B) 1 (C) 3 (D) infinite

10. The set of values of x in (−π, π) satisfying $|4\sin x - 1| < \sqrt{5}$ is given by -

(A) $\left(-\frac{9\pi}{10}, -\frac{\pi}{5}\right)$ (B) $\left(-\frac{\pi}{5}, \frac{\pi}{10}\right)$ (C) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (D) $\left(\frac{3\pi}{10}, \frac{7\pi}{10}\right)$

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Section–I(ii) Multiple Correct Type : (Maximum Marks: 24)

11. $\sqrt{\cos 2x} + \sqrt{1+\sin 2x} = 2\sqrt{\sin x + \cos x}$, $(n \in I)$ if -

(A) $\sin x + \cos x = 0$ (B) $x = 2n\pi$ (C) $x = n\pi - \frac{\pi}{4}$ (D) none of these

12. The solution set of the equation $|\tan x| = \cot x$ is/are -

(A) $2n\pi + \frac{5\pi}{4}$ (B) $n\pi \pm \frac{\pi}{4}$ (C) $2n\pi + \frac{\pi}{4}$ (D) none of these

13. Values of θ satisfying the equation $(\sqrt{3}-1)\cos\theta + (\sqrt{3}+1)\sin\theta = (\sqrt{3}-1)$ is :-

(A) $2n\pi + \frac{5\pi}{6}$ (B) $2n\pi - \frac{5\pi}{6}$ (C) $2n\pi$ (D) $2n\pi \pm \frac{5\pi}{12}$

14. If $x = \sin(\alpha - \beta).\sin(\gamma - \delta)$, $y = \sin(\beta - \gamma).\sin(\alpha - \delta)$ and $z = \sin(\gamma - \alpha).\sin(\beta - \delta)$, then -

(A) $x + y + z = 0$ (B) $x + y - z = 0$ (C) $y + z - x = 0$ (D) $x^3 + y^3 + z^3 = 3xyz$

15. The equation $\log_{x+1}(x - 0.5) = \log_{x-0.5}(x+1)$ has

(A) no real solution (B) no prime solution

(C) an irrational solution (D) no composite solution

16. Which of the following statements does not hold good ?

(A) $\log_{10}(1.4^2 - 1)$ is positive

(B) the equation $\log_a(a+2) = 2$ is satisfied by two integral values of a

(C) $\log_{0.1}\cot\frac{3\pi}{8}$ is negative

(D) if $m = 4^{\log_4 7}$ and $n = \left(\frac{1}{9}\right)^{-2\log_3 7}$ then $n = m^4$

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Section–II (iii) : Integer Type : (Maximum Marks: 24)

1. Let triangle ABC be the isosceles triangle and $\angle A = 100°$, if $\sin A + \sin B + \sin C = x$ and $\cos A + \cos B + \cos C = y$, then $(x^2 + y^2)$ is

2. If $a^x = b$, $b^y = c$, $c^z = a$ (where a,b,c, x,y,z are positive real numbers), then the value of $\frac{\sum_{r=1}^{89} \log_{10}\tan\left(\frac{\pi r}{180}\right)}{xyz}$ is equal to

3. If $\ell n\left(\frac{a+b}{3}\right) = \left(\frac{\ell na + \ell nb}{2}\right)$, then $\frac{a}{b} + \frac{b}{a}$ is equal to -

4. Number of solution of the equation $\cos^4 x - \sin^{36} x = 1$ in $[0, 2\pi]$

5. The number of values of x lying in the interval $(-\pi, \pi)$ which satisfy the equation $81^{(1+|\sin x|+|\sin^2 x|+|\sin^3 x|+.....\infty)} = (3)^8$

6. Let $\lambda_1\lambda_2 \in [0,\pi]$ are the solutions of the equation $\operatorname{cosec}\left(\frac{\pi}{4}+x\right) + \operatorname{cosec}\left(\frac{\pi}{4}-x\right) = 2\sqrt{2}$, then $8(\sin^2\lambda_1 + \sin^2\lambda_2)$ is equal to

MATHS (Sets, Relation & Statistics) GUIDED REVISION # 20

TIME : 60 Min. **M.M. : 81**

Section–I(i) Single Correct Type (Maximum Marks:57)

1. If A, B are two sets then $(A - B) \cup (B - A) \cup (\bar{A} \cap \bar{B})$ is equal to

(A) $\bar{A} \cup \bar{B}$

(B) $\overline{(A \cup B)}$

(C) ϕ

(D) U (where U is universal set and A & B are subsets of U)

2. The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{1344}$, then the standard deviation of the second sample is :

(A) 8 (B) 6

(C) 4 (D) 5

3. The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is:

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(A) (11, 26) (B) (10.5, 25)

(C) (11, 25) (D) (10.5, 26)

4. Two finite sets A and B have 'm' and 'n' elements respectively. The total number of subsets of A is 56 more then the total number of subsets of B. The value of (m + n) is-

(A) 13 (B) 9

(C) 10 (D) 15

5. Consider three observations a, b and c such that b = a + c. If the standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true?

(A) $b^2 = 3(a^2 + c^2) + 9d^2$

(B) $b^2 = a^2 + c^2 + 3d^2$

(C) $b^2 = 3(a^2 + c^2 + d^2)$

(D) $b^2 = 3(a^2 + c^2) - 9d^2$

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6. Let in a series of 2n observations, half of them are equal to a and remaining half are equal to $-a$. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20, respectively. Then the value of $a^2 + b^2$ is equal to :

(A) 425 (B) 650

(C) 250 (D) 925

7. Let M be a relation from R (Set of real number) to R defined by M = {(a, b) | a, b $\in$ R and a + b is an irrational number}, then relation M is -

(A) reflexive and symmetric only (B) symmetric only

(C) symmetric and transitive only (D) equivalence relation

8. Let S = {(x, y)|siny = sinx, x, y $\in$ R}, then S is

(A) not transitive relation (B) equivalence relation

(C) transitive but not reflexive relation (D) partial order relation

9. Let R be a relation defined on a set of real numbers such that a R b and if $|a| \leq b$, then relation R is :-

(A) Reflexive (B) Symmetric

(C) Transitive (D) Equivalence

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10. $x_i (i = 1,2,3,.....n)$ denotes a distribution whose variance is 10 then variance of a distribution $3x_1 + 2, 3x_2 + 2, 3x_3 + 2,.....3x_n + 2$ will be-

(A) 10 (B) 30 (C) $\frac{10}{3}$ (D) 90

11. For 21 distinct observations, if observations greater than the median are increased by 2, then identify the correct statement-

(A) Mean and median are unchanged.

(B) Both median and mean are changed.

(C) Median is changed, mean is same.

(D) Median is same, mean is changed.

12. The arithmetic mean of n observations is $\bar{x}$. If each of these observations is divided by α and increased by β, mean of new series will be -

(A) $\frac{\bar{x} + n\beta}{n}$ (B) $\frac{\bar{x} + \alpha\beta}{\alpha}$ (C) $\frac{\bar{x} + \alpha\beta}{n}$ (D) $\frac{\bar{x} + \alpha\beta}{n\beta}$

13. Let $a_1, a_2, \ldots\ldots a_{101}$ is a group of real numbers such that $a_i > a_{i+1}$ for all values of i and mean square deviation of the group is minimum about the number 'a_{51}', then mode of the group will be -

(A) $2a_{51}$ (B) a_{51}

(C) a_{50} (D) a_{52}

14. If mean deviations about median of x, 2x, 3x, 4x, 5x, 6x, 7x, 8x, 9x, 10x is 30, then |x| equals -

(A) 12 (B) 11

(C) 10 (D) 9

15. Mean and variance of a set of 6 terms is 11 and 24 respectively and the mean and variance of another set of 3 terms is 14 and 36 respectively. Then variance of all 9 terms is equal to -

(A) 40 (B) 30

(C) 50 (D) 35

16. Let $x_1, x_2, \ldots\ldots, x_{100}$ are 100 observations such that $\sum x_i = 0$, $\sum_{1 \leqslant i < j \leqslant 100} |x_i x_j| = 80000$ & mean deviation from their mean is 5, then their standard deviation, is-

(A) 10 (B) 30

(C) 40 (D) 50

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17. The A.M. of a set of 50 numbers is 38. If two numbers of the set namely 55 and 45 are discarded, the A.M. of the remaining set of numbers is

(A) 36 (B) 36.5

(C) 37.5 (D) 38.5

18. let $x_1, x_2 \ldots\ldots x_n$ be n observations such that $\Sigma x_i^2 = 200$ and $\Sigma x_i = 40$ then a possible value of n among the following is

(A) 4 (B) 5

(C) 6 (D) 10

19. The arithmetic mean of a set of observation is $\bar{X}$. If each observation is divided by α and then is increased by 10, then the mean of the new series is

(A) $\frac{\bar{X}}{\alpha}$ (B) $\frac{\bar{X} + 10}{\alpha}$

(C) $\frac{\bar{X} + 10\alpha}{\alpha}$ (D) $\alpha\bar{X} + 10$

Section–II (iii) : Integer Type : (Maximum Marks: 24)

1. A relation R is defined on set A = {a,b,c} such that R ≡ {(a,a), (b,b),(b,c),(c,b)}, then minimum number of additional elements required in R so that R becomes an equivalance relation, is

2. Let the mean and variance of four numbers 3, 7, x and y(x > y) be 5 and 10 respectively. Then the mean of four numbers 3 + 2x, 7 + 2y, x + y and x − y is _____.

3. Let $X_1, X_2, ..., X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is _____.

4. The variance of 10 observations is 16. If each observation is doubled, then standard deviation of new data will be-

5. In a given frequency distribution, the respective values of mean and median are 21 and 22. The value of mode is

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6. If mean deviation of sample 1, 2, 3, x about 3 is 1 (where x > 3), then possible value of variance is